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NaVILA: Legged Robot Vision-Language-Action Model for Navigation

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https://navila-bot.github.io

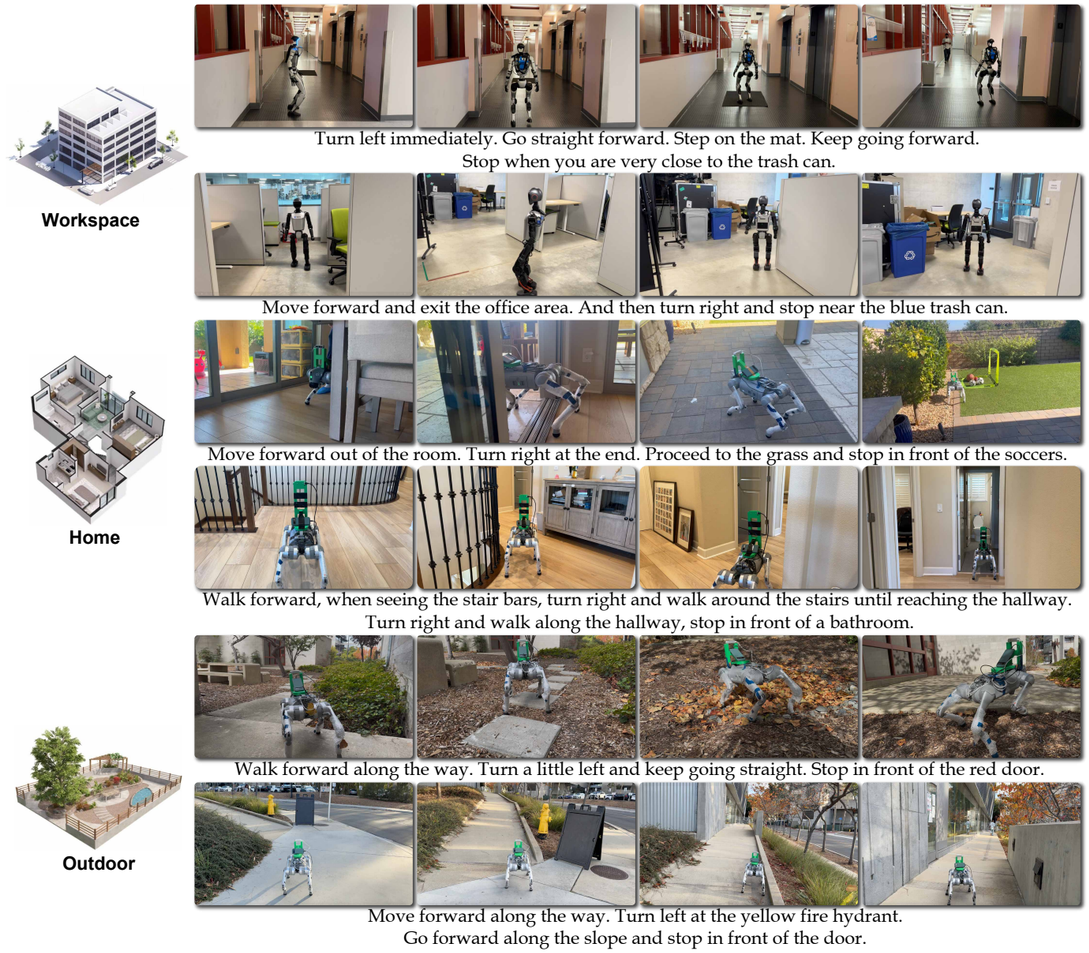

Fig. 1: Real-world demonstration of NaVILA: Upon receiving human instructions, NaVILA uses a vision-language model to process RGB video frames and employs locomotion skills to execute the task on a robot. The robot successfully handles long-horizon navigation tasks and operates safely in challenging environments.

***Abstract*—This paper proposes to solve the problem of Vision-and-Language Navigation with legged robots, which not only provides a flexible way for humans to command but also allows the robot to navigate through more challenging and cluttered scenes. However, it is non-trivial to translate human language instructions all the way to low-level leg joint actions. We propose NaVILA, a 2-level framework that unifies a Vision-Language-Action model (VLA) with locomotion skills. Instead of directly predicting low-level actions from VLA, NaVILA first generates mid-level actions with spatial information in the form of language, (e.g., "moving forward 75cm"), which serves as an input for a visual locomotion RL policy for execution. NaVILA substantially**

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NaVILA：用于导航的腿式机器人视觉-语言-动作模型

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图 1：NaVILA 的真实世界演示：在接收到人类指令后，NaVILA 使用视觉语言模型处理 RGB 视频帧，并运用运动技能在机器人上执行任务。该机器人成功处理长时程导航任务，并能在具有挑战性的环境中安全运行。

摘要——本文旨在解决腿式机器人的视觉与语言导航问题，这不仅为人类提供了一种灵活的下达指令方式，还使机器人能够在更具挑战性和更杂乱的场景中导航。然而，翻译人类语言并非易事

∗同等贡献，按字母顺序排列。 † 同等指导。

improves previous approaches on existing benchmarks. The same advantages are demonstrated in our newly developed benchmarks with IsaacLab, featuring more realistic scenes, low-level controls, and real-world robot experiments.

I. Introduction

The ability to perform Vision-and-Language Navigation (VLN) has become a foundational component in modern robotics systems. With VLN, a robot is expected to navigate around unseen environments without a provided map following a language instruction [1–6]. This not only offers a better interface for humans, but also strengthen cross-scene generalization through languages. In this paper, we further extend the study of VLN with legged robots (e.g., quadruped or humanoid). Using legs instead of wheels allows robots to navigate in more challenging and cluttered scenarios. As the examples shown in Fig. 1, our robot can navigate through a messy laboratory space with narrow walkways, transition from room to room in a house, as well as tackle outdoor challenging environments such as uneven terrains with small rocks, holes, and troughs.

To translate language to action, the robot needs to reason about the input language, and perform closed-loop planning as well as low-level control. With the recent advancement in Large Language Models (LLMs) and Vision-Language Models (VLMs), several end-to-end Vision-Language-Action (VLA) systems have been developed [7–9]. These systems fine-tune a general-propose VLM with large-scale robot manipulation demonstrations to produce low-level actions. While unifying reasoning and execution in a single model is fascinating and shows encouraging results, it is worth diving deeper into the question: Is there a better way to represent actions beyond the quantized low-level commands? After all, LLMs and VLMs were primarily trained with natural language. Unifying reasoning and execution becomes challenging when we need to convert that reasoning into precise, non-verbal actions.

Inspired by the recent progress on VLM [10, 11] for spatial location and distance reasoning, we propose **NaVILA**, a two-level framework for legged robot VLN: A VLM is fine-tuned to output a **mid-level action** (VLA) in the form of language such as "turn right 30 degrees", and a low-level visual locomotion policy is trained to follow this instruction for execution. The mid-level action output of the VLA conveys the location and direction information without the low-level commands. The advantages of this framework are three-fold: (i) By decoupling low-level execution from VLAs, the same VLA can be applied across different robots by swapping the low-level policy; (ii) Representing actions as mid-level language instructions enables VLA training with diverse data sources, including real human videos and reasoning QA tasks. This enhances reasoning capabilities without overfitting outputs to specific low-level commands and can leverage real-world data for generalization; (iii) NaVILA operates on two distinct timescales: the VLA, typically a large and computationally intensive model, runs at a lower frequency, providing high-level navigation commands; while the locomotion policy operates in real-time. This dual-frequency approach allows the locomotion policy to handle sophisticated obstacle avoidance and increases overall robustness.

To train the VLA, we demonstrate how to (i) integrate historical context and current observations in VLN within existing VLM frameworks, (ii) create a specialized navigation prompt tailored for VLN tasks, (iii) utilize real-world data from YouTube human touring videos to improve navigation in continuous environments, and (iv) introduce a carefully curated dataset blend designed to enhance VLN generalizability. These strategies allow us to fine-tune a general-purpose image-based VLM into a navigation-focused agent while simultaneously training it on general vision-language datasets, thereby maintaining its broad generalization capabilities. Moreover, this is the first work to show that direct training on human videos improves navigation in continuous environments.

To train robust locomotion skills, we employ a single-stage approach to learn vision-based locomotion policy. We construct a height map from raw LiDAR point clouds and introduce randomization to bridge the sim-to-real gap. This controller takes the output from our VLA model, converts it into command velocities, and tracks these velocities by controlling the positions of the joints. This end-to-end approach enables the training of visual locomotion skills that are both robust and safe, facilitating deployment in real-world, challenging environments (e.g., strong sunlight or near certain transparent surfaces).

In our experiments, we show that our VLA significantly outperforms the state-of-the-arts on classic VLN benchmarks, with over 17% improvement in success rate. Additionally, our single-stage locomotion policy outperforms previous policy distillation-based methods by a substantial margin. To better simulate the challenges of locomotion navigation in VLN, we introduce a new benchmark, VLN-CE-Isaac, using Isaac Sim. This benchmark considers detailed robotic joint movements and interactions with environments, which prior VLN works have not explored. In our VLN-CE-Isaac experiments, our vision-based policy outperforms the blind policy by a significant margin, showing a 14% improvement in success rate. We also demonstrate that our VLA can be deployed across different robots (Unitree Go2, Unitree H1, Booster T1), each using distinct locomotion skills. Finally, we deploy NaVILA in the real world, exhibiting impressive robustness and achieving an 88% success rate on 25 instructions, including a 75% success rate on complex instructions across diverse scenes.

II. Method

NaVILA integrates high-level visual language understanding with low-level locomotion control (Fig.2). It employs a VLM to process single-view images and generate waypoint instructions in natural language, which a locomotion policy translates into precise joint movements for real-time robot control. The synergy between the VLM's reasoning and the locomotion policy's execution enables NaVILA to generalize across diverse environments. We first describe how we tame VLMs for high-level VLN in Sec.II-A, then outline our robot configuration and locomotion policy in Sec. II-B.

在现有基准上提升了先前方法的表现。同样的优势也在我们新开发的基于 IsaacLab 的基准测试中得到展示，这些基准具有更逼真的场景、底层控制以及真实世界机器人实验。

I. 引言

执行视觉语言导航（Vision-and-Language Navigation, VLN）的能力已成为现代机器人系统的一项基础组成部分。借助 VLN，机器人应能在没有提供地图的情况下，遵循语言指令在未见过的环境中导航 [1–6]。这不仅为人类提供了更好的交互界面，还通过语言增强了跨场景泛化能力。在本文中，我们进一步将 VLN 研究扩展到腿式机器人（例如四足或人形机器人）。使用腿而非轮子使机器人能够在更具挑战性和更杂乱的环境中导航。如图 1所示，我们的机器人可以穿行于带有狭窄通道的杂乱实验室空间，在房屋中从一个房间移动到另一个房间，也能应对具有挑战性的户外环境，例如散布着小石块、坑洞和沟槽的不平坦地形。

要将语言转化为动作，机器人需要推理输入语言，并执行闭环规划以及底层控制。随着大语言模型（LLMs）和视觉语言模型（VLMs）的近期进展，若干端到端视觉语言动作（VLA）系统已被开发出来 [7–9]。这些系统使用大规模机器人操作演示对通用型 VLM 进行微调，以生成底层动作。虽然将推理与执行统一到单个模型中令人着迷，并展现出令人鼓舞的结果，但值得深入探讨一个问题：除了量化的底层命令之外，是否有更好的动作表示方式？毕竟，LLMs 和 VLMs 主要使用自然语言进行训练。当我们需要将这种推理转化为精确的非语言动作时，统一推理与执行就变得具有挑战性。

受 VLM [10, 11] 在空间位置和距离推理方面近期进展的启发，我们提出了 **NaVILA**，一种用于足式机器人 VLN 的两级框架：对 VLM 进行微调，使其以语言形式输出 **中层动作**（VLA），例如“向右转 30 度”；同时训练一个低层视觉运动策略来按照该指令执行。VLA 的中层动作输出无需低层命令，就能传达位置和方向信息。该框架的优势有三方面：(i) 通过将低层执行与 VLA 解耦，只需替换低层策略，同一个 VLA 就能应用于不同的机器人；(ii) 将动作表示为中层语言指令，使 VLA 能够利用包括真实人类视频和推理问答任务在内的多样化数据源进行训练。这增强了推理能力，又不会使输出过度拟合特定的低层命令，并且能够利用真实世界数据进行泛化；(iii) NaVILA 在两个不同的时间尺度上运行：VLA 通常是一个大型且计算密集的模型，以较低频率运行，提供高层导航命令；而运动策略则实时运行。这种双频方法使得运动策略能够处理复杂的避障，并提高整体鲁棒性。

为了训练 VLA，我们展示了如何 (i) 在现有的 VLM 框架内整合 VLN 中的历史上下文和当前观测，(ii) 为 VLN 任务创建专门的导航提示，(iii) 利用来自 YouTube 人类游览视频的真实世界数据来改进连续环境中的导航，以及 (iv) 引入精心策划的数据集混合，旨在增强 VLN 的泛化能力。这些策略使我们能够将通用的基于图像的 VLM 微调为专注于导航的智能体，同时在其上训练通用的视觉-语言数据集，从而保持其广泛的泛化能力。此外，这是首个表明在人类视频上直接训练可以改进连续环境中导航的工作。

为了训练鲁棒的运动技能，我们采用单阶段方法来学习基于视觉的运动策略。我们从原始 LiDAR 点云构建高度图，并引入随机化来弥合仿真到现实的差距。该控制器接收来自我们 VLA 模型的输出，将其转换为命令速度，并通过控制关节位置来跟踪这些速度。这种端到端方法能够训练既鲁棒又安全的视觉运动技能，便于部署在真实世界的挑战性环境中（例如，强烈的阳光或靠近某些透明表面）。

在我们的实验中，我们表明我们的 VLA 在经典 VLN 基准上显著优于当前最先进方法，成功率提升超过 17%。此外，我们的单阶段运动策略大幅优于先前基于策略蒸馏的方法。为了更好地模拟 VLN 中运动导航的挑战，我们使用 Isaac Sim 引入了一个新基准 VLN-CE-Isaac。该基准考虑了精细的机器人关节运动以及与环境交互，而此前的 VLN 工作尚未探索这些内容。在我们的 VLN-CE-Isaac 实验中，我们基于视觉的策略显著优于盲策略，成功率提升了 14%。我们还证明，我们的 VLA 可以部署到不同机器人上（Unitree Go2、Unitree H1、Booster T1），每台机器人使用不同的运动技能。最后，我们将 NaVILA 部署到真实世界，展现出令人印象深刻的鲁棒性，并在 25 条指令上达到了 88% 的成功率，其中包括在不同场景下复杂指令上达到 75% 的成功率。

II. 方法

NaVILA 将高级视觉语言理解与低级运动控制相结合（图2）。它利用 VLM 处理单视图图像并生成自然语言形式的航点指令，然后由运动策略将其转换为精确的关节动作，以实现实时机器人控制。VLM 推理与运动策略执行之间的协同作用，使 NaVILA 能够泛化到多样化环境中。我们首先描述如何让 VLM 胜任高级 VLN，见第 II-A节，然后概述我们的机器人配置和运动策略，见第 II-B节。

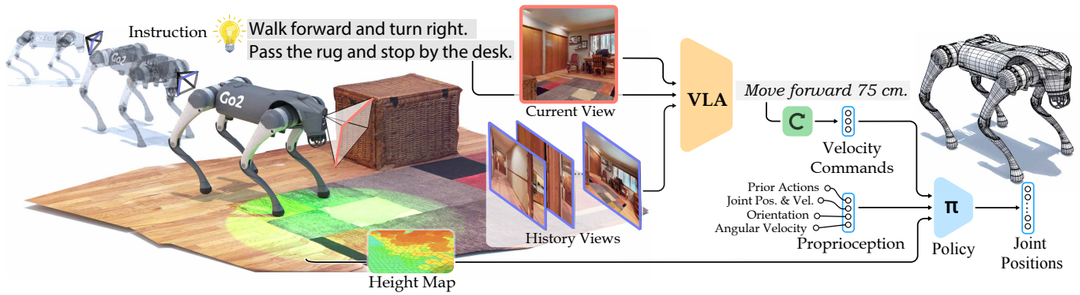

Fig. 2: NaVILA is a two-level framework combining high-level visual language understanding with low-level locomotion control. Our VLA model processes single-view images to produce mid-level actions in natural language, which are then converted into precise joint movements by an advanced low-level locomotion policy. This integration allows for strong generalization and adaptability across different real-world environments, and can operate the robot in real-time.

A. *Taming VLMs for Vision Language Navigation*

VLN requires processing video inputs as observations. A common approach to handling video inputs in VLMs is through video encoders [12]. However, recent progress in VLMs has largely been driven by the availability of image-text data. While there have been efforts to extend this success to video encoders, the lack of large, high-quality video-text datasets has limited their pre-training. To address this challenge, we opt for image-based vision-language models in our approach. These models exhibit stronger generalization abilities and possess broader knowledge, making them more suitable for tackling the generalization challenges in VLN. Specifically, we built our approach upon VILA [13–19], a family of efficient VLMs for both understanding and generation. VILA's pre-training has proven particularly effective for multi-image reasoning, making it especially suitable for VLN tasks where understanding sequential image relationships is critical.

VILA Preliminary. VILA consists of three main components: a vision encoder, a projector, and an LLM. The vision encoder processes the input images, converting them into a sequence of visual tokens. These tokens are then downsampled and mapped into the language domain via an MLP projector. Afterward, the projected tokens, along with text tokens, are sent to the LLM for auto-regressive generation. When handling videos, VILA uniformly sampled frames at regular intervals. It puts all the frame information before any text. A typical prompt for describing a video might look like "⟨frame3⟩⟨frame6⟩⟨frame9⟩...Tell me about this video." Notably, with sequence parallel training [16], VILA can include frames up to 1024. VILA undergoes a 3-stage training process: first, it pre-trains a connector between the frozen LLM and vision backbones using alignment data [20]; then it pre-trains both the connector and the LLM using text-image interleaved corpus [21, 22]; and finally, it fine-tunes all modules (vision encoder, connector, LLM) with instruction tuning data [20, 23].

Navigation Prompts. In vision-language navigation tasks, images from different time steps serve two distinct purposes. The image at time step t represents the current observation, which is crucial for a VLN agent to make immediate decisions (e.g., turning right at an intersection or stopping when the goal is reached). On the other hand, frames before time step t are historical frames that function as a memory bank, helping the agent track overall progress (e.g., remembering the starting location, reasoning about places already visited and planning the next step). Uniformly sampling frames at regular intervals, as done in VILA, is not ideal because it doesn't differentiate between these two types of representations. Therefore, we first extract the most recent frame t as the current observation and then uniformly sample frames from the preceding $t-1$ frames, ensuring the first frame is always included. Additionally, since current and historical observations serve different roles, we distinguish them in our task prompt using textual cues like `a video of historical observations`: for memory frames and `current observation`: for the latest frame. Unlike [12], we avoid introducing additional special tokens that could complicate the LLM's learning process. Instead, we adhere to our design principle of keeping both the input and output of LLM in the language domain to fully leverage the reasoning capabilities of the pre-trained LLM. By integrating these tokens for historical and current observations with the navigation instruction, we construct a navigation task prompt, as shown in Fig. 2.

Learning from Human Videos. Recent studies [24–26] have shown that collecting trajectory-instruction pairs from human videos can enhance navigation capabilities. However, prior work has been limited to discrete navigation settings and has mainly used real videos for pre-training to reduce domain gaps or improve landmark understanding, rather than for directly training navigation models. Extending this approach to continuous settings presents a significant challenge due to the difficulty of obtaining continuous action labels. Recent advances in metric-pose estimation in the wild have now made this feasible, enabling us to extract spatial understanding from human videos and train navigation models directly.

Our data pipeline, shown in Fig. 4, starts with 2K egocentric touring videos from YouTube, which provide a rich source of real-world data to learn robot navigation from human behavior. We process these videos into 20K diverse and representative

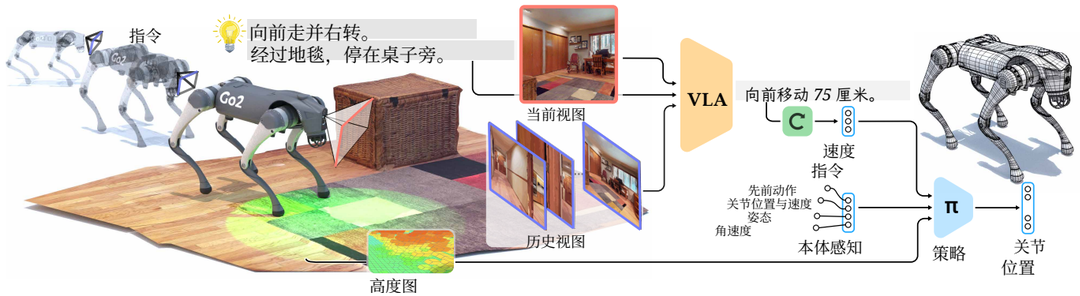

图 2：NaVILA 是一个双层框架，将高层视觉语言理解与底层运动控制相结合。我们的 VLA 模型处理单视图图像，以生成自然语言形式的中层动作，随后由先进的底层运动策略转换为精确的关节运动。这种集成能够在不同的真实环境中实现强大的泛化能力和适应性，并可实时操控机器人。

A. 驯服*VLM*以用于视觉语言导航

VLN 需要将视频输入作为观测进行处理。在 VLM 中处理视频输入的一种常见方法是通过视频编码器 [12]。然而，VLM 的近期进展在很大程度上是由图像-文本数据的可用性推动的。尽管已有工作试图将这一成功扩展到视频编码器，但缺乏大规模、高质量的视频-文本数据集限制了它们的预训练。为应对这一挑战，我们的方法选择基于图像的视觉语言模型。这些模型表现出更强的泛化能力，并拥有更广泛的知识，因而更适合应对 VLN 中的泛化挑战。具体来说，我们将方法构建在 VILA [13–19],一个高效 VLM 系列，兼顾理解与生成。VILA 的预训练已被证明对多图像推理尤为有效，这使其特别适合 VLN 任务，因为理解序列图像之间的关系在这些任务中至关重要。

VILA 预备知识。VILA 由三个主要组件构成：视觉编码器、投影器和 LLM。视觉编码器处理输入图像，将其转换为一系列视觉 token。随后，这些 token 被下采样，并通过 MLP 投影器映射到语言域。之后，投影后的 token 与文本 token 一起被送入 LLM 进行自回归生成。在处理视频时，VILA 以固定间隔均匀采样帧。它将所有帧信息置于任何文本之前。用于描述视频的典型提示可能如下："⟨帧3⟩⟨帧6⟩⟨帧9⟩...告诉我这段视频的内容。"值得注意的是，借助序列并行训练 [16]，VILA 最多可包含 1024 帧。VILA 经历 3 个阶段的训练过程：首先，它使用对齐数据预训练冻结 LLM 与视觉骨干网络之间的连接器 [20]；然后，它使用文本-图像交错语料库预训练连接器和 LLM [21, 22]；最后，它使用指令微调数据微调所有模块（视觉编码器、连接器、LLM） [20, 23]。

导航提示词。在视觉语言导航任务中，来自不同时间步的图像具有两种截然不同的用途。时间步 t 的图像表示当前观测，这对于 VLN 智能体做出即时决策至关重要（例如，在路口右转，或在到达目标时停止）。另一方面，时间步 t 之前的帧是历史帧，起到记忆库的作用，帮助智能体跟踪整体进展（例如，记住起始位置，推理已访问过的地点并规划下一步）。像 VILA 那样按固定间隔均匀采样帧并不理想，因为它没有区分这两类表示。因此，我们首先提取最近的帧 t 作为当前观测，然后从之前的 $t-1$ 帧中均匀采样，确保始终包含第一帧。此外，由于当前观测和历史观测扮演不同角色，我们在任务提示中使用文本线索来区分它们，例如 历史观测视频：用于记忆帧，而 当前观测：用于最新帧。与 [12],不同，我们避免引入可能使 LLM 学习过程复杂化的额外特殊标记。相反，我们遵循自己的设计原则，即将 LLM 的输入和输出都保持在语言领域，以充分利用预训练 LLM 的推理能力。通过将用于历史观测和当前观测的这些标记与导航指令整合，我们构建了一个导航任务提示，如图 2所示。

从人类视频中学习。 近期研究 [24–26] 表明，从人类视频中收集轨迹-指令对可以增强导航能力。然而，先前工作仅限于离散导航场景，且主要使用真实视频进行预训练，以缩小领域差距或改进地标理解，而非直接训练导航模型。将这一方法扩展到连续场景面临重大挑战，因为难以获得连续动作标签。近期野外度量位姿估计的进展已使这成为可能，这使我们能够从人类视频中提取空间理解，并直接训练导航模型。

我们的数据管道，如图 4 所示，始于来自 YouTube 的 2K 段第一视角游览视频，这些视频提供了丰富的真实世界数据，可用于从人类行为中学习机器人导航。我们将这些视频处理成 20K 条多样且具有代表性的

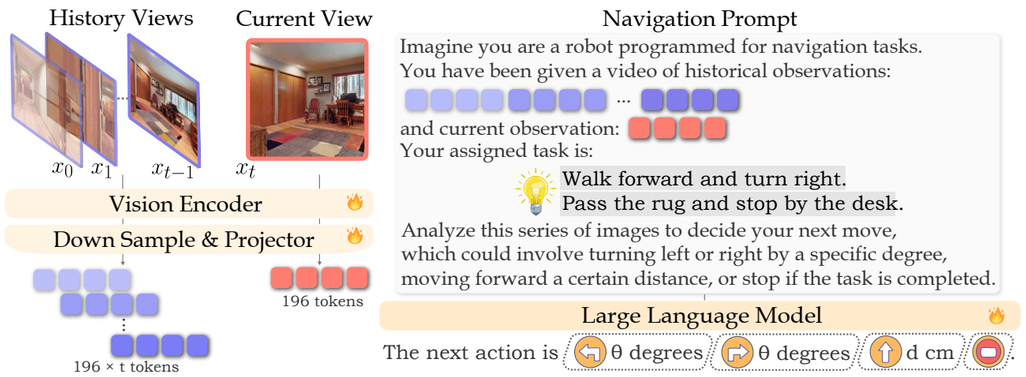

Fig. 3: Overview of our VLA framework. We denote the purple blocks (■) as memory tokens sampled from historical frames, and the red blocks (■) as the current observation tokens. 🔥 denotes trainable parameters. In our experiments, we tested configurations with 8 to 64 frames for t.

trajectories using entropy-based sampling [26]. Next, we estimate camera poses using MASt3R [27] to extract step-by-step actions, and we generate natural language instructions for each trajectory using VLM-based [13] captioning followed by LLM [28] rephrasing. This approach allows us to leverage human demonstrations for continuous navigation, a capability that was previously non-trivial to achieve.

Supervised Fine-tuning Data Blend. Effective Supervised Fine-tuning (SFT) data is crucial for developing a robust vision-language action model. The model should specialize in embodied tasks while avoiding overfitting to specific actions. It should also generalize well to real-world scenarios while retaining broad-world knowledge. Thanks to NaVILA's modular framework, which offers exceptional scalability and adaptability, integrating diverse data sources into our pipeline is straightforward. This flexibility allows us to enhance generalizability for navigation. Our SFT data blend is designed from four perspectives: (1) Navigational data from real videos, (2) Navigational data from simulations, (3) Auxiliary navigational data, and (4) General VQA datasets.

For simulated navigational data, the available VLN datasets in continuous environments are limited, with only R2R-CE [29] and RxR-CE [30] providing sparse path points converted from discrete VLN versions. We leverage both datasets within the Habitat simulator, using a shortest path follower to generate action sequences along the geodesic shortest path. This results in step-wise navigation videos, where each sample comprises a $(t+1)$-frame video and the corresponding oracle action at time step t. To encourage the LLM to generate continuous value labels for distances and angles, we merge consecutive actions (e.g., combining two forward 25 cm steps into a single forward 50 cm step), with a maximum of three consecutive actions. This merging process not only reduces dataset size for more efficient processing but also introduces greater diversity in actions, mitigating overfitting. Additionally, to address label imbalance—particularly the underrepresentation of the stop action—we apply a rebalancing technique for a more even distribution. All navigation-specific data undergo the previously described frame extraction strategy and are paired with navigation task prompts.

To further improve scene understanding and address the limited instructions in R2R-CE and RxR-CE, we incorporate auxiliary navigational datasets. Following [12], we use augmented instructions from EnvDrop [31] and introduce an auxiliary task of navigation trajectory summarization. Given a trajectory video, we sample frames by retaining the first frame and uniformly selecting historical frames, using the annotated instructions as labels. The LLM is then tasked with describing the robot's trajectory based on these frames. To further enhance spatial scene understanding, we integrate the ScanQA [32] dataset, which features real-world 3D scan QA pairs with human-edited questions and free-form answers grounded in 3D objects. For training, we use multi-view RGB images from the raw scans to support this task.

Finally, to maintain the model's general capabilities, we incorporate general VQA datasets from [23, 33, 34]. This comprehensive dataset design ensures that NaVILA can generalize effectively to novel scenes and real-world environments.

Training and Inference Paradigm. Our training process begins with the stage two model of VILA, which has already undergone visual language corpus pre-training. We then apply our SFT data blend to train the entire VLM for one epoch, following standard practices. During this training, all three components—vision encoder, connector, and LLM—are unfrozen. For the inference phase, we implement a regular expression parser [35], to extract action types (such as forward or turn left) and their corresponding arguments (like specific distance or angles) from the LLM output. This method has demonstrated effectiveness in both simulated environments and real-world experiments, where we empirically found that all actions throughout all experiments are successfully matched and mapped.

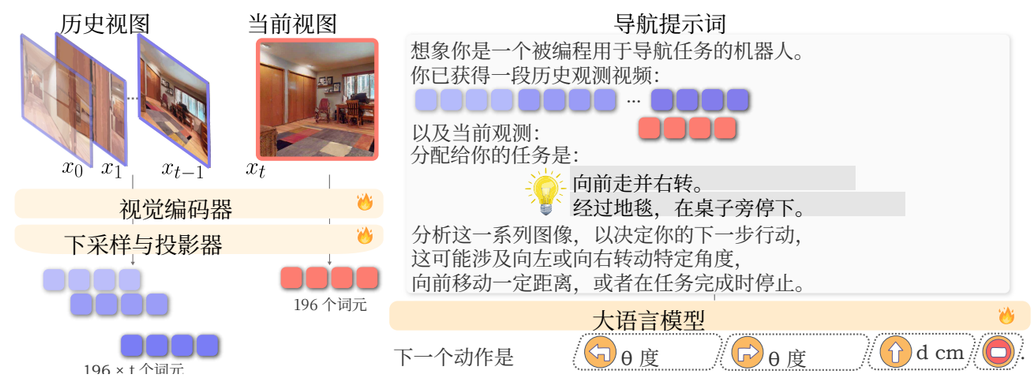

图 3：我们的 VLA 框架概览。我们将紫色块（■）表示为从历史帧采样的记忆词元，将红色块（■）表示为当前观测词元。表示可训练参数。🔥在我们的实验中，我们测试了针对 🔥 使用 8 到 64 帧的配置。

轨迹，使用基于熵的采样 [26]。接下来，我们使用 MASt3R 估计相机位姿 [27] 以提取逐步动作，并且我们为每条轨迹生成自然语言指令，使用基于 VLM 的 [13] 描述生成，随后由 LLM [28]改写。这种方法使我们能够利用人类演示进行连续导航，而这一能力此前难以实现。

监督微调数据混合。有效的监督微调（SFT）数据对于开发稳健的视觉-语言动作模型至关重要。模型应专注于具身任务，同时避免对特定动作过拟合。它还应在保留广泛世界知识的同时，很好地泛化到真实世界场景。得益于 NaVILA 的模块化框架，该框架具有出色的可扩展性和适应性，将多样化的数据源集成到我们的流水线中非常简单。这种灵活性使我们能够增强导航的泛化能力。我们的 SFT 数据混合方案从四个角度设计：（1）来自真实视频的导航数据，（2）来自仿真的导航数据，（3）辅助导航数据，以及（4）通用 VQA 数据集。

对于模拟导航数据，连续环境中可用的 VLN 数据集有限，只有 R2R-CE [29] 和 RxR-CE [30] 提供了从离散 VLN 版本转换而来的稀疏路径点。我们在 Habitat 模拟器中利用这两个数据集，使用最短路径跟随器沿测地最短路径生成动作序列。这会生成逐步导航视频，其中每个样本包含一个 $t+1$ 帧视频以及时间步 t 对应的 oracle 动作。为了鼓励 LLM 为距离和角度生成连续值标签，我们合并连续动作（例如，将两个向前 25 cm 的步合并为一个向前 50 cm 的步），最多合并三个连续动作。这一合并过程不仅缩小了数据集规模以实现更高效的处理，还引入了更大的动作多样性，从而缓解过拟合。此外，为解决标签不平衡——尤其是停止动作表示不足的问题——我们应用了一种重平衡技术

以实现更均匀的分布。所有导航专用数据都经过前述的帧提取策略处理，并与导航任务提示配对。

为了进一步提升场景理解能力，并弥补 R2R-CE 和 RxR-CE 中指令数量有限的问题，我们引入了辅助导航数据集。遵循 [12]，，我们使用来自 EnvDrop 的增强指令 [31]，并引入一项导航轨迹摘要的辅助任务。给定一段轨迹视频，我们通过保留第一帧并均匀选取历史帧来进行采样，并将标注指令作为标签。随后，LLM 的任务是根据这些帧描述机器人的轨迹。为了进一步增强空间场景理解，我们整合了 ScanQA [32] 数据集，该数据集包含真实世界的 3D 扫描问答对，其中的问题由人工编辑，答案则是基于 3D 物体的自由形式回答。在训练中，我们使用来自原始扫描的多视角 RGB 图像来支持这一任务。

最后，为了保持模型的通用能力，我们引入了来自 [23, 33, 34] 的通用 VQA 数据集。这种全面的数据集设计确保 NaVILA 能够有效泛化到新场景和真实世界环境中。

训练与推理范式。我们的训练过程从 VILA 的第二阶段模型开始，该模型已经过视觉语言语料库预训练。随后，我们遵循标准做法，应用 SFT 数据混合集对整个 VLM 训练一个 epoch。在此训练过程中，视觉编码器、连接器和 LLM 这三个组件均不冻结。在推理阶段，我们实现了一个正则表达式解析器 [35]，，用于从 LLM 输出中提取动作类型（例如前进或左转）及其对应参数（如特定距离或角度）。该方法已在模拟环境和真实世界实验中展现出有效性；我们通过经验发现，所有实验中的所有动作都能成功匹配和映射。

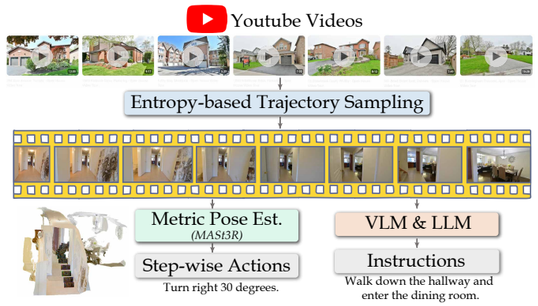

Fig. 4: Data pipeline for transforming human touring videos in the wild into pairwise navigation data within a continuous environment. We begin by processing the videos into meaningful trajectories through entropy-based sampling [26]. Then we extract step-wise actions through metric camera pose estimation [27], and utilize VLM [13] and LLM [28] to generate instructions.

B. Visual Locomotion Policy

In this section, we begin with a brief overview of the Go2 robot dog, the experimental platform used in this work. Next, we describe the development of the end-to-end vision-based control policy, which interprets high-level language navigation commands from the VLM and converts them into precise joint movements. This control policy is trained in the Isaac Sim simulator using Isaac Lab [36] and then directly deployed to the real-world robot.

Go2 Robot. As shown in Fig. 5, the robot is equipped with a LiDAR sensor mounted at the base of its head, broadcasting point clouds at a frequency of 15Hz. The robot features 18 degrees of freedom (DoFs), comprising 6 DoFs for its base and 3 DoFs for each of its four legs. In the policy training process, we left the 6 DoFs on the base unconstrained so that the policy only controls the 12 joint motors on the legs.

Interpreting High-level Commands. As in our formulation, VLM outputs a fixed set of actionable words, such as {move forward, turn left, turn right, stop}, we casts these instructions to fixed command velocities $\{0.5\,\mathrm{m\,s^{-1}},\ \frac{\pi}{6}\,\mathrm{rad\,s^{-1}},\ -\frac{\pi}{6}\,\mathrm{rad\,s^{-1}},\ 0\}$ and execute with corresponding time durations to align with the specific VLM value.

Low-level Action and Observation Space. The action space $\mathbf{a}$ of the control policy is defined as the desired joint position $q^d \in \mathbb{R}^{12}$, which is converted into torque input for the simulator using the stiffness and dampness. We adopt the PPO algorithm [37] to train the policy. During training, the critic observes the privileged environment and generates a value function to update the actor, while the actor only receives sensor data available in the real world. The observation space of the critic $\mathbf{o}^c$ contains the proprioception and velocity command at the current time step t and a privileged terrain height scan around the robot. The proprioceptive data includes robot linear and angular velocity, orientation, joint positions, joint velocities, and the previous action. In the actor's observation space, $\mathbf{o}^a$, linear velocity is excluded, as it is unavailable in the real world, and instead, a history of proprioceptive data is used to infer this information implicitly. The robot perceives the surrounding terrain using a height map from the LiDAR sensor.

(a) Simulation (b) Real

Fig. 5: Height map reconstruction from point cloud. (**a**) Go2 robot follows velocity commands while avoiding obstacles in simulation. Red dots show LiDAR points raycasting from the sensor center to the terrain mesh. The right image shows a preprocessed height map with values clipped to sensor constraints; darker colors indicate higher heights. (**b**) Safe locomotion near glass. The top-down height map detects glass surfaces where depth and RGB images fail.

Incorporating Height Map from LiDAR Point Cloud. Given LiDAR's superior ability to detect transparent objects and robust performance under strong sunlight, we chose the manufacturer-provided LiDAR as the primary sensor for perceiving the robot's surroundings and ensuring safe navigation. The Unitree L1 generates point clouds with a wide field of view of $360° \times 90°$, from which we create a 2.5D height map based on the parameters listed in the Supplementary. For each voxel grid, the lowest value within the range is selected, and a maximum filter is then applied over the last 5 lidar point clouds to smooth the resulting height map.

Training. Different from most existing works [38–41] that utilize the two-stage teacher-student training paradigm, we adopt a single-stage manner to train the locomotion policy. Compared to two-stage training, single-stage RL is more time-efficient as it eliminates the need for policy distillation. Additionally, the policy interacts directly with the environment, allowing it to explore and potentially discover novel strategies. With the support of ray-casting in Isaac Lab, our vision-based RL policy training achieves a high throughput over 60K FPS on an RTX 4090 GPU.

III. Experiments

We conduct experiments to answer the following questions: (1) How does our VLA's performance compare to state-of-the-art methods in VLN-CE benchmarks and general spatial scene understanding tasks? (Sec. III-A) (2) How does the performance of our single-stage visual locomotion policy compare to policy distillation-based approaches? (Sec. III-B) (3) How to evaluate locomotion navigation in simulators, and how effective and flexible is NaVILA in these scenarios? (Sec. III-C) (4) Can NaVILA pipeline be successfully deployed in real robot VLN experiments? (Sec. III-D)

A. High-level VLA Performance

VLN-CE Benchmarks. We evaluate our VLA on the VLN-CE benchmarks, which provide continuous environments for executing navigational actions in reconstructed photorealistic indoor scenes. We focus on the val-unseen split in both R2R (Room-to-Room) and RxR (Room-across-Room) datasets

图4：用于将真实场景中的人类游览视频转换为连续环境中的成对导航数据的数据流水线。我们首先通过基于熵的采样将这些视频处理为有意义的轨迹 [26]。然后，我们通过度量相机位姿估计提取逐步动作 [27]，并利用VLM [13] 和LLM [28]生成指令。

B. 视觉运动策略

本节中，我们首先简要介绍 Go2 机器狗，这是本工作所使用的实验平台。接下来，我们描述端到端基于视觉的控制策略的开发，该策略解释来自 VLM 的高级语言导航指令，并将其转换为精确的关节运动。该控制策略在 Isaac Sim 模拟器中使用 Isaac Lab [36] 进行训练，然后直接部署到真实世界机器人上。

Go2 机器人。如图5所示，机器人头部基座处安装有一个 LiDAR 传感器，以 15Hz 的频率广播点云。该机器人具有 18 个自由度（DoFs），包括基座的 6 个自由度和四条腿各 3 个自由度。在策略训练过程中，我们让基座上的 6 个自由度不受约束，因此策略仅控制腿部的 12 个关节电机。

解释高层指令。按照我们的形式化设定，VLM 输出一组固定的可执行词，例如 {前进、左转、右转、停止}。我们将这些指令转换为固定指令速度 $\{0.5\,\mathrm{m/s^{-1}}$、$\frac{\pi}{6}\,\mathrm{rad/s^{-1}}$、$-\frac{\pi}{6}\,\mathrm{rad/s^{-1}}$、$0\}$，并以相应的持续时间执行，以与具体的 VLM 值对齐。

底层动作与观测空间。控制策略的动作空间$\mathbf{a}$ 被定义为期望关节位置$q^d \in \mathbb{R}^{12}$，其通过刚度和阻尼转换为仿真器的力矩输入。我们采用 PPO 算法 [37] 来训练策略。在训练期间，critic 观察特权环境并生成价值函数以更新 actor，而 actor 仅接收真实世界中可用的传感器数据。critic 的观测空间 $\mathbf{o}^c$ 包含当前时间步的本体感觉和速度指令 t 以及机器人周围的特权地形高度扫描。本体感觉数据包括机器人的线速度和角速度、姿态、关节位置、关节速度以及上一动作。在 actor 的观测空间中，$\mathbf{o}^a$，线速度被排除，因为它在

现实世界，相反，利用本体感知数据的历史来隐式推断这一信息。机器人感知

(a) 仿真 (b) 真实

图 5：从点云重建高度图。(**a**) 在仿真中，Go2 机器人遵循速度指令并避开障碍物。红点表示从传感器中心向地形网格进行射线投射的 LiDAR 点。右图展示了预处理后的高度图，其数值被裁剪到传感器约束范围内；颜色越深表示高度越高。(**b**) 玻璃附近的安全运动。俯视高度图可检测到深度与 RGB 图像无法检测到的玻璃表面。

周围地形，使用来自 LiDAR 传感器的高度图。

利用 LiDAR 点云构建高度图。鉴于 LiDAR 在检测透明物体方面具有卓越能力，并且在强烈阳光下性能稳健，我们选择厂商提供的 LiDAR 作为主要传感器，用于感知机器人周围环境并确保安全导航。Unitree L1 生成具有 $360° \times 90°$ 宽视场的点云，我们基于补充材料中列出的参数，由此创建 2.5D 高度图。对于每个体素网格，选择该范围内的最低值，然后在最近 5 帧激光雷达点云上应用最大值滤波，以平滑得到的高度图。

训练。与大多数现有工作 [38–41] 采用两阶段教师-学生训练范式不同，我们采用单阶段方式来训练运动策略。与两阶段训练相比，单阶段强化学习更节省时间，因为它无需策略蒸馏。此外，策略直接与环境交互，使其能够探索并可能发现新颖策略。在 Isaac Lab 中射线投射的支持下，我们基于视觉的强化学习策略训练在 RTX 4090 GPU 上实现了超过 60K FPS 的高吞吐量。

III. 实验

我们进行实验以回答以下问题：（1）我们的 VLA 在 VLN-CE 基准和一般空间场景理解任务上的性能与最先进方法相比如何？（第 III-A 节）（2）我们的单阶段视觉运动策略与基于策略蒸馏的方法相比性能如何？（第 III-B 节）（3）如何在模拟器中评估运动导航，以及 NaVILA 在这些场景中多么有效和灵活？（第 III-C 节）（4）NaVILA 流程能否成功部署在真实机器人 VLN 实验中？（第 III-D 节）

A. 高层 *VLA* 性能

VLN-CE 基准测试。我们在 VLN-CE 基准测试上评估我们的 VLA，这些基准测试为在重建的逼真室内场景中执行导航动作提供了连续环境。我们重点关注 R2R（Room-to-Room）和 RxR（Room-across-Room）数据集中的 val-unseen 划分。

TABLE I: Comparison with state-of-the-art methods on the Val-Unseen split of R2R-CE [29] and RxR-CE [30]. * indicates methods using the waypoint predictor from Hong et al. [42]. NaVILA outperforms all methods that do not rely on simulator pre-trained waypoint predictors, even when those methods leverage additional inputs such as depth, panoramic views, and odometry.

	Observation				R2R Val-Unseen				RxR Val-Unseen			
	S.RGB	Pano.	Depth	Odo.	NE ↓	OS ↑	SR ↑	SPL ↑	NE ↓	SR ↑	SPL ↑	nDTW ↑
HPN+DN* [43]		✓	✓	✓	6.31	40.0	36.0	34.0	-	-	-	-
CMA* [42]		✓	✓	✓	6.20	52.0	41.0	36.0	8.76	26.5	22.1	47.0
VLN⟳BERT* [42]		✓	✓	✓	5.74	53.0	44.0	39.0	8.98	27.0	22.6	46.7
Sim2Sim* [44]		✓	✓	✓	6.07	52.0	43.0	36.0	-	-	-	-
GridMM* [45]		✓	✓	✓	5.11	61.0	49.0	41.0	-	-	-	-
Ego^2-Map* [46]		✓	✓	✓	5.54	56.0	47.0	41.0	-	-	-	-
DreamWalker* [47]		✓	✓	✓	5.53	59.0	49.0	44.0	-	-	-	-
Reborn* [48]		✓	✓	✓	5.40	57.0	50.0	46.0	5.98	48.6	42.0	63.3
ETPNav* [49]		✓	✓	✓	4.71	65.0	57.0	49.0	5.64	54.7	44.8	61.9
HNR* [50]		✓	✓	✓	4.42	67.0	61.0	51.0	5.50	56.3	46.7	63.5
BEVBert* [51]		✓	✓	✓	4.57	67.0	59.0	50.0	4.00	68.5	-	69.6
HAMT+ScaleVLN* [52]		✓	✓	✓	4.80	-	55.0	51.0	-	-	-	-
AG-CMTP [53]		✓	✓	✓	7.90	39.0	23.0	19.0	-	-	-	-
R2R-CMTP [53]		✓	✓	✓	7.90	38.0	26.0	22.0	-	-	-	-
LAW [54]	✓		✓	✓	6.83	44.0	35.0	31.0	10.90	8.0	8.0	38.0
CM2 [55]	✓		✓	✓	7.02	41.0	34.0	27.0	-	-	-	-
WS-MGMap [56]	✓		✓	✓	6.28	47.0	38.0	34.0	-	-	-	-
AO-Planner [57]		✓	✓		5.55	59.0	47.0	33.0	7.06	43.3	30.5	50.1
Seq2Seq [58]	✓		✓		7.77	37.0	25.0	22.0	12.10	13.9	11.9	30.8
CMA [58]	✓		✓		7.37	40.0	32.0	30.0	-	-	-	-
RGB-Seq2Seq [58]	✓				10.10	8.0	0.0	0.0	-	-	-	-
RGB-CMA [58]	✓				9.55	10.0	5.0	4.0	-	-	-	-
NaVid [12]	✓				5.47	49.0	37.0	35.0	-	-	-	-
NaVILA	✓				**5.22**	**62.5**	**54.0**	**49.0**	**6.77**	**49.3**	**44.0**	**58.8**

TABLE II: Cross-dataset performance on the RxR-CE [30] Val-Unseen split. All results are obtained without training on the RxR-CE training set. NaVILA significantly outperforms NaVid [12], the current single-view state-of-the-art.

	Observation			RxR Val-Unseen			
	S.RGB	Depth	Odo.	NE ↓	OS ↑	SR ↑	SPL ↑
LAW [54]	✓	✓	✓	10.87	21.0	8.0	8.0
CM2 [55]	✓	✓	✓	8.98	25.3	14.4	9.2
WS-MGMap [56]	✓	✓	✓	9.83	29.8	15.0	12.1
Seq2Seq [58]	✓	✓		11.8	5.02	3.51	3.43
CMA [58]	✓	✓		11.7	10.7	4.41	2.47
RGB-Seq2Seq [12]	✓			11.2	12.2	0.0	0.0
RGB-CMA [12]	✓			9.55	14.8	0.0	0.0
A^{2}NAV [59]	✓			-	-	16.8	6.3
NaVid [12]	✓			**8.41**	34.5	23.8	21.2
NaVILA	✓			8.78	**46.8**	**34.3**	**28.2**

within VLN-CE, as these are the two most recognized benchmarks in VLN. We employ the following widely used evaluation metrics for VLN tasks: Navigation Error (NE), Oracle Success Rate (OS), Success Rate (SR), Success-weighted Path Length (SPL), and normalize dynamic time wrapping (nDTW). We show results in Table I, where NaVILA significantly surpasses all baselines that do not rely on simulator pre-trained waypoint predictors in both benchmarks using a single model. Notably, this also marks the first time a VLN agent, trained solely on single-view RGB input, achieves comparable or superior results to models that use panoramic views, odometry, or simulator-pretrained waypoint predictors. This suggests that NaVILA's strong generalization capabilities can effectively compensate for the limited observations in RGB views or sensors.

To evaluate the cross-dataset performance, we follow [12] by training NaVILA exclusively on R2R samples, while leaving out the RxR training set. We then evaluate its zero-shot performance on the RxR Val-Unseen split. As shown in Table II, our method significantly outperforms NaVid, the current state-of-the-art model, with a substantial 10% improvement in SR.

Spatial Scene Understanding Benchmarks. As a general navigation agent, robust spatial scene understanding (e.g., object localization, referring, and spatial reasoning) is crucial. To evaluate NaVILA's capabilities in scene understanding, we conduct evaluations on the ScanQA Validation benchmark, a widely used dataset for 3D Question Answering. ScanQA is based on real-world scans, and we use multi-view images from these scans as input to query NaVILA for answers. As shown in Table III, NaVILA significantly outperforms the previous state-of-the-art model, NaviLLM [60], by a substantial margin (20 points higher on the CIDEr score). Moreover, when using 64 frames, NaVILA's performance demonstrates superior performance compared to state-of-the-art 3D-based large multi-modal models [61, 62]. This is particularly noteworthy as these other models require either 3D scans or RGBD data with camera poses as inputs, while our method achieves better results with less observation.

B. Low-level RL Policy Performance

To highlight the advantages of our RL policy over policy distillation-based approaches, we compared it to Regularized Online Adaptation (ROA) [68]. In ROA training, the model first learns a privileged encoder that processes height scan points and other privileged observations. This privileged encoder then supervises an adaptation encoder, which takes the same 2.5D heightmap as our low-level policy as input. We evaluated both approaches using three metrics: linear velocity

表 I：在 R2R-CE 的 Val-Unseen 划分上与最先进方法的比较 [29] 和 RxR-CE [30]。 * 表示使用 Hong 等人提出的路点预测器的方法。 [42]。NaVILA 优于所有不依赖模拟器预训练路点预测器的方法，即使这些方法利用了深度、全景视图和里程计等额外输入。

	观测				R2R Val-Unseen				RxR Val-Unseen			
	单目 RGB	全景	深度	Odo.	NE ↓	OS ↑	SR ↑	SPL ↑	NE ↓	SR ↑	SPL ↑	nDTW ↑
HPN+DN* [43]		✓	✓	✓	6.31	40.0	36.0	34.0	-	-	-	-
CMA* [42]		✓	✓	✓	6.20	52.0	41.0	36.0	8.76	26.5	22.1	47.0
VLN⟳BERT* [42]		✓	✓	✓	5.74	53.0	44.0	39.0	8.98	27.0	22.6	46.7
Sim2Sim* [44]		✓	✓	✓	6.07	52.0	43.0	36.0	-	-	-	-
GridMM* [45]		✓	✓	✓	5.11	61.0	49.0	41.0	-	-	-	-
Ego^2-Map* [46]		✓	✓	✓	5.54	56.0	47.0	41.0	-	-	-	-
DreamWalker* [47]		✓	✓	✓	5.53	59.0	49.0	44.0	-	-	-	-
Reborn* [48]		✓	✓	✓	5.40	57.0	50.0	46.0	5.98	48.6	42.0	63.3
ETPNav* [49]		✓	✓	✓	4.71	65.0	57.0	49.0	5.64	54.7	44.8	61.9
HNR* [50]		✓	✓	✓	4.42	67.0	61.0	51.0	5.50	56.3	46.7	63.5
BEVBert* [51]		✓	✓	✓	4.57	67.0	59.0	50.0	4.00	68.5	-	69.6
HAMT+ScaleVLN* [52]		✓	✓	✓	4.80	-	55.0	51.0	-	-	-	-
AG-CMTP [53]		✓	✓	✓	7.90	39.0	23.0	19.0	-	-	-	-
R2R-CMTP [53]		✓	✓	✓	7.90	38.0	26.0	22.0	-	-	-	-
LAW [54]	✓		✓	✓	6.83	44.0	35.0	31.0	10.90	8.0	8.0	38.0
CM2 [55]	✓		✓	✓	7.02	41.0	34.0	27.0	-	-	-	-
WS-MGMap [56]	✓		✓	✓	6.28	47.0	38.0	34.0	-	-	-	-
AO-Planner [57]		✓	✓		5.55	59.0	47.0	33.0	7.06	43.3	30.5	50.1
Seq2Seq [58]	✓		✓		7.77	37.0	25.0	22.0	12.10	13.9	11.9	30.8
CMA [58]	✓		✓		7.37	40.0	32.0	30.0	-	-	-	-
RGB-Seq2Seq [58]	✓				10.10	8.0	0.0	0.0	-	-	-	-
RGB-CMA [58]	✓				9.55	10.0	5.0	4.0	-	-	-	-
NaVid [12]	✓				5.47	49.0	37.0	35.0	-	-	-	-
NaVILA	✓				**5.22**	**62.5**	**54.0**	**49.0**	**6.77**	**49.3**	**44.0**	**58.8**

表 II：在 RxR-CE [30] Val-Unseen 划分上的跨数据集性能。所有结果均是在未使用 RxR-CE 训练集训练的情况下获得的。NaVILA 显著优于 NaVid [12]，（当前单视图的最先进方法）。

	观测			RxR 验证-未见			
	S.RGB	深度	里程计.	NE ↓	OS ↑	SR ↑	SPL ↑
LAW [54]	✓	✓	✓	10.87	21.0	8.0	8.0
CM2 [55]	✓	✓	✓	8.98	25.3	14.4	9.2
WS-MGMap [56]	✓	✓	✓	9.83	29.8	15.0	12.1
Seq2Seq [58]	✓	✓		11.8	5.02	3.51	3.43
CMA [58]	✓	✓		11.7	10.7	4.41	2.47
RGB-Seq2Seq [12]	✓			11.2	12.2	0.0	0.0
RGB-CMA [12]	✓			9.55	14.8	0.0	0.0
A^{2}NAV [59]	✓			-	-	16.8	6.3
NaVid [12]	✓			**8.41**	34.5	23.8	21.2
NaVILA	✓			8.78			**28.2**

在 VLN-CE 中，因为它们是 VLN 中最受认可的两个基准。我们采用以下广泛用于 VLN 任务的评估指标：导航误差（NE）、Oracle 成功率（OS）、成功率（SR）、成功加权路径长度（SPL）以及归一化动态时间规整（nDTW）。我们将结果显示在表 I 中，其中 NaVILA 使用单一模型，在两个基准上都显著超越了所有不依赖模拟器预训练路点预测器的基线。值得注意的是，这也是首次有 VLN 智能体仅基于单视图 RGB 输入进行训练，就能取得与使用全景视图、里程计或模拟器预训练路点预测器的模型相当甚至更优的结果。这表明 NaVILA 强大的泛化能力可以有效弥补 RGB 视图或传感器中观测信息有限的问题。

为了评估跨数据集性能，我们遵循 [12]，通过仅在 R2R 样本上训练 NaVILA，而排除 RxR 训练集。然后，我们在 RxR Val-Unseen 划分上评估其零样本性能。如表 II 所示，我们的方法显著优于当前最先进的模型 NaVid，在 SR 上大幅提升了 10%。

空间场景理解基准。作为通用导航智能体，稳健的空间场景理解（例如物体定位、指代和空间推理）至关重要。为了评估 NaVILA 在场景理解方面的能力，我们在 ScanQA 验证基准上进行评估，这是一个广泛用于 3D 问答的数据集。ScanQA 基于真实世界扫描，我们使用这些扫描的多视图图像作为输入，向 NaVILA 查询答案。如表III 所示，NaVILA 显著优于此前的最先进模型 NaviLLM [60]，，提升幅度很大（CIDEr 分数高 20 分）。此外，当使用 64 帧时，NaVILA 的表现相比最先进的基于 3D 的大型多模态模型展现出更优性能 [61, 62]。这一点尤其值得注意，因为这些其他模型需要 3D 扫描或带有相机位姿的 RGBD 数据作为输入，而我们的方法以更少的观测取得了更好的结果。

B. 底层强化学习策略性能

为了突出我们的强化学习策略相较于基于策略蒸馏方法的优势，我们将其与正则化在线自适应（ROA） [68]进行了比较。在 ROA 训练中，模型首先学习一个特权编码器，用于处理高度扫描点和其他特权观测。这个特权编码器随后监督一个自适应编码器，该编码器以与我们的低层策略相同的 2.5D 高度图作为输入。我们使用三个指标评估了这两种方法：线速度

TABLE III: Evaluation of spatial scene understanding performance on the ScanQA dataset [32] Validation split. NaVILA outperforms current state-of-the-art VLA models and demonstrates superior performance to other 3D LMMs that require additional input, such as depth or camera pose. Note that $*$ indicates 3D LMMs that require task-specific fine-tuning on the ScanQA dataset.

	ScanQA Validation				
	Bleu-4 ↑	Rouge ↑	Cider ↑	Meteor ↑	EM ↑
Task-specific Specialist					
VoteNet+MCAN [63]	6.2	29.8	54.7	11.4	17.3
ScanRefer+MCAN [63]	7.9	30.0	55.4	11.5	18.6
ScanQA [32]	10.1	33.3	64.9	13.1	21.0
3D-VisTA [64]	10.4	35.7	69.6	13.9	22.4
3D Large Multi-modal Models					
3D-LLM$_{(flamingo)}$* [65]	7.2	32.3	59.2	12.2	20.4
3D-LLM$_{(BLIP2-flant5)}$* [65]	12.0	35.7	69.4	14.5	20.5
LL3DA* [66]	13.5	37.3	76.8	15.9	-
Chat-3Dv2* [67]	14.0	-	87.6	-	-
Scene-LLM* [61]	12.0	40.0	80.0	16.6	27.2
LEO [62]	13.2	49.2	101.4	20.0	24.5
2D Vision-Langauge-Action Models					
NaviLLM [60]	12.0	38.4	75.9	15.4	23.0
NaVILA (8 frames)	14.8	46.4	95.1	18.7	27.0
NaVILA (64 frames)	**16.9**	**49.3**	**102.7**	**20.1**	**28.6**

error, angular velocity error, and collision rate. The first two metrics assess how accurately the policy follows velocity commands, while the third measures the model's obstacle avoidance capabilities. As shown in Table V, our low-level policy outperforms ROA in all three metrics, particularly achieving a significantly lower collision rate, demonstrating the effectiveness of our training approach.

C. Legged Robot Navigation Performance in Simulation

High-fidelity VLN-CE-Isaac Benchmark. Currently, there are no VLN-CE benchmarks tailored specifically for legged robots. Existing benchmarks [29, 30] for vision-language navigation are based on the Habitat [69] simulator, which focuses on high-level planning without addressing precise low-level robotic control. For instance, agents in Habitat can navigate through narrow gaps, such as a 10 cm space between two sofas, which is impractical for legged robots like quadrupeds or humanoids. To overcome this limitation, we introduce a new benchmark VLN-CE-Isaac built on Isaac Sim. Isaac Sim's high-fidelity simulation captures detailed robotic joint movements and interactions with the environment, enabling comprehensive evaluations of the entire navigation pipeline, from high-level planning to precise robotic execution. We incorporate the same scenes from R2R, with robots deployed in the environment, as shown in Fig. 6. From the 1,839 trajectories in the R2R Val-Unseen split, we select 1,077 traversable trajectories with high-quality meshes to ensure realistic navigation scenarios. For consistency, we evaluate performance using the same metrics as prior work.

Notably, VLN-CE-Isaac is compatible with a variety of robotic platforms. To demonstrate this flexibility, we test our NaVILA model on a Unitree Go2 robot and also a Unitree H1 robot within the benchmark. To highlight the effectiveness of the vision-based policy, we compare it against a proprioception-only (*blind*) policy. As shown in Table IV,

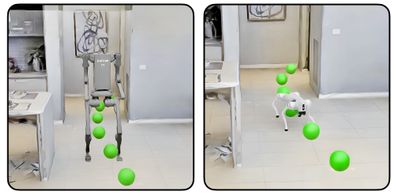

Fig. 6: VLN-CE-Isaac Benchmark visualization.

TABLE IV: VLN-CE-Isaac evaluation results.

	Low-level Observation			VLN-CE-Isaac			
	Proprio.	LiDAR	Height Scan	NE ↓	OS ↑	SR ↑	SPL ↑
Oracle				*5.25*	*59.8*	*51.3*	*46.9*
Unitree Go2							
NaVILA-Blind	✓			6.03	49.0	36.2	33.3
NaVILA-Vision	✓	✓		**5.49**	**58.7**	**50.2**	**45.5**
Unitree H1							
NaVILA-Blind	✓			7.67	33.3	24.4	21.0
NaVILA-Vision	✓		✓	**5.86**	**54.6**	**45.3**	**40.3**

TABLE V: Low level policy performance.

	Linear Vel. Error ↓	Angular Vel. Error ↓	Collision Rate ↓
ROA$_{(w/BCLoss)}$ [68]	0.189	0.152	3.25
ROA [68]	0.161	0.152	3.09
NaVILA	**0.066**	**0.113**	**0.81**

the vision-based policy outperforms the blind policy by 14% in Success Rate in Go2 settings and 21% in H1 settings, owing to its superior obstacle avoidance capability. We also compare NaVILAs with a baseline using Oracle's low-level policy (assuming perfect command execution without realistic physics). Results show a 15% lower success rate on the Go2 setup and a 27% lower success rate on the H1 setup when Oracle policy is not presented. These performance gaps highlight the increased challenges and realism introduced by our benchmark. Additionally, we also observe that the success rate of NaVILA on the H1 robot is significantly lower than on the Go2, which is expected due to the larger size of the humanoid robot.

D. Real World Evaluation

We then conduct experiments in the real world, using 25 instructions, each repeated three times, covering both simple and complex tasks across three types of environments: `Workspace`, `Home`, and `Outdoor` open environments. Simple instructions consist of one or two navigation commands, where the robot does not need to navigate between rooms (e.g., "Go to the chair and stop"). In contrast, complex instructions involve three or more commands, requiring the robot to traverse multiple rooms or landmarks (e.g., "Walk out of the room, turn right, enter the room in front of you, and stop at the table"). We use standard metrics (SR and NE) and compare NaVILA against GPT-4o, a state-of-the-art VLM known for its strong generalizability. As shown in Table VI, NaVILA significantly outperforms GPT-4o in both SR and NE. We also ablate the effectiveness of adding human videos, and the

表 III：在 ScanQA 数据集上的空间场景理解性能评估 [32] 验证集。NaVILA 优于当前最先进的 VLA 模型，并表现出优于其他需要额外输入（如深度或相机位姿）的 3D LMM 的性能。注意，$*$ 表示需要在 ScanQA 数据集上进行任务特定微调的 3D LMM。

	ScanQA 验证集				
	Bleu-4 ↑	Rouge ↑	Cider ↑	Meteor ↑	EM ↑
任务专用专家					
VoteNet+MCAN [63]	6.2	29.8	54.7	11.4	17.3
ScanRefer+MCAN [63]	7.9	30.0	55.4	11.5	18.6
ScanQA [32]	10.1	33.3	64.9	13.1	21.0
3D-VisTA [64]	10.4	35.7	69.6	13.9	22.4
3D大型多模态模型					
3D-LLM$_{(flamingo)}$* [65]	7.2	32.3	59.2	12.2	20.4
3D-LLM$_{(BLIP2-flant5)}$* [65]	12.0	35.7	69.4	14.5	20.5
LL3DA* [66]	13.5	37.3	76.8	15.9	-
Chat-3Dv2* [67]	14.0	-	87.6	-	-
Scene-LLM* [61]	12.0	40.0	80.0	16.6	27.2
LEO [62]	13.2	49.2	101.4	20.0	24.5
2D 视觉-语言-动作模型					
NaviLLM [60]	12.0	38.4	75.9	15.4	23.0
NaVILA （8帧）	14.8	46.4	95.1	18.7	27.0
NaVILA （64 帧）	**16.9**	**49.3**	**102.7**	**20.1**	**28.6**

误差、角速度误差和碰撞率。前两个指标评估策略遵循速度命令的准确程度，而第三个指标衡量模型的避障能力。如表V所示，我们的低层策略在全部三个指标上均优于 ROA，尤其是实现了显著更低的碰撞率，证明了我们训练方法的有效性。

C. 仿真中的腿式机器人导航性能

高保真 VLN-CE-Isaac 基准测试。目前，尚无专门为腿式机器人定制的 VLN-CE 基准测试。现有基准测试 [29, 30]针对视觉-语言导航，均基于 Habitat [69] 模拟器，其侧重于高层规划，而未涉及精确的低层机器人控制。例如，Habitat 中的智能体可以穿过狭窄间隙，比如两个沙发之间 10 厘米的空间，这对于四足或人形等腿式机器人来说并不现实。为克服这一局限，我们提出了一个基于 Isaac Sim 构建的新基准测试 VLN-CE-Isaac。Isaac Sim 的高保真仿真能够捕捉详细的机器人关节运动以及与环境的交互，从而支持对整个导航流程进行全面评估，从高层规划到精确的机器人执行。我们采用了与 R2R 相同的场景，并将机器人部署在环境中，如图 6 所示。从 R2R Val-Unseen 划分的 1,839 条轨迹中，我们选择 1,077 条具有高质量网格的可通行轨迹，以确保导航场景逼真。为保持一致，我们使用与先前工作相同的指标来评估性能。

值得注意的是，VLN-CE-Isaac 与多种机器人平台兼容。为了展示这种灵活性，我们在基准测试中，在 Unitree Go2 机器人以及 Unitree H1 机器人上测试了我们的 NaVILA 模型。为了突出基于视觉的策略的有效性，我们将其与仅依赖本体感知的（盲）策略。如表 IV，

图 6：VLN-CE-Isaac 基准可视化。

表 IV：VLN-CE-Isaac 评估结果。

	低层观测			VLN-CE-Isaac			
	本体感知	LiDAR	高度扫描	NE ↓	OS ↑	SR ↑	SPL ↑
Oracle				*5.25*	*59.8*		*46.9*
宇树 Go2							
NaVILA-Blind	✓			6.03	49.0	36.2	33.3
NaVILA-Vision	✓	✓		**5.49**			**45.5**
宇树 H1							
NaVILA-盲	✓			7.67	33.3	24.4	21.0
NaVILA-视觉	✓		✓	**5.86**			**40.3**

表 V：底层策略性能。

	线速度误差 ↓	角速度误差 ↓	碰撞速率 ↓
ROA$_{(w/BCLoss)}$ [68]	0.189	0.152	3.25
ROA [68]	0.161	0.152	3.09
NaVILA	**0.066**	**0.113**	**0.81**

基于视觉的策略在 Go2 设置中的成功率比盲策略高 14%，在 H1 设置中高 21%，这得益于其更强的避障能力。我们还将 NaVILA 与使用 Oracle 底层策略的基线进行了比较（假设命令执行完美，且不考虑真实物理）。结果表明，在不使用 Oracle 策略时，Go2 设置的成功率低 15%，H1 设置的成功率低 27%。这些性能差距凸显了我们的基准测试引入的更大挑战和真实感。此外，我们还观察到，NaVILA 在 H1 机器人上的成功率显著低于在 Go2 上的成功率，这在意料之中，因为人形机器人的尺寸更大。

D. 真实世界评估

随后我们在真实世界中进行实验，使用 25 条指令，每条重复三次，覆盖三类环境中的简单和复杂任务：工作区、家庭以及户外开放环境。简单指令由一到两条导航命令组成，机器人无需在房间之间导航（例如，"走到椅子那里并停下"）。相比之下，复杂指令包含三条或更多命令，要求机器人穿过多个房间或地标（例如，"走出房间，右转，进入你面前的房间，并在桌子旁停下"）。我们使用标准指标（SR 和 NE），并将 NaVILA 与 GPT-4o 进行比较，后者是以强大泛化能力著称的最先进的 VLM。如表VI所示，NaVILA 在 SR 和 NE 两项指标上均显著优于 GPT-4o。我们还消融了添加人类视频的有效性，并且

TABLE VI: Real-world experiments on quadruped (Unitree Go2) and humanoid (Booster T1) conducted in different environments (`Workspace`, `Home`, and `Outdoor`). Simple and Complex refer to simple and complex instruction-following tasks, respectively. Note that † indicates models trained without human touring videos.

	Workspace				Home				Outdoor			
	Simple		Complex		Simple		Complex		Simple		Complex	
	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑
Unitree Go2												
GPT-4o [28]	2.01	0.67	2.38	0.33	1.49	0.53	3.00	0.00	-	0.67	-	0.50
NaVILA †	2.00	0.60	1.81	0.73	2.17	0.47	2.32	0.40	-	0.00	-	0.00
NaVILA	**1.29**	**1.00**	**1.76**	**0.80**	**1.15**	**1.00**	**1.76**	**0.67**	-	**1.00**	-	**0.83**
Booster T1												
GPT-4o [28]	1.53	0.67	2.78	0.13	-	-	-	-	-	0.44	-	-
NaVILA †	2.36	0.40	2.16	0.62	-	-	-	-	-	0.22	-	-
NaVILA	**1.18**	**0.93**	**1.91**	**0.67**	-	-	-	-	-	**0.89**	-	-

results show that with the help of human videos, the model can generalize better to outdoor scenes and achieve higher success rates across all environments. To demonstrate the flexibility of our two-level approach, we also evaluated it on a Booster Dynamics T1 humanoid robot, using the same VLA model without any retraining. Despite variations such as changes in camera height and camera view angle, NaVILA consistently outperformes the baselines, highlighting the strong generalization capabilities of our model. Our qualitative results are presented in Fig.1 and Fig.7. In Fig. 7, we demonstrate integration with speech recognition, enabling voice-controlled navigation through our framework. These results highlight the effectiveness of NaVILA in bridging the gap between vision-language understanding and real-world navigation tasks.

IV. Related Work

Visual Navigation. Visual navigation has been a long-standing research topic in robotics [71–74]. Classical methods rely on pre-computed [75] or geometric maps built with depth sensors [76] or monocular cameras while localizing the robot (SLAM) [77, 78]. More recently, learning-based approaches using Imitation Learning [79, 80] and Reinforcement Learning [81, 82] have demonstrated strong performance and expanded applications to Vision-Language Navigation.

Vision-Language Navigation. Vision-Language Navigation (VLN) is a fundamental challenge in embodied AI, where agents navigate complex environments using visual cues and natural language instructions. The field has evolved significantly over time. Early research [1, 30, 83] focused on discrete navigation in simulated environments like MP3D [84], where agents teleport between predefined nodes on a navigation graph [31, 85–91]. As foundation models advanced, many VLN systems improved dramatically by leveraging large-scale pre-trained models [25, 92] and pre-training techniques [24, 52, 93], approaching human-level performance in this setting. However, this setup emphasized high-level decision-making while neglecting the challenges of underlying motion control. Recently, research [12, 54–57] has shifted towards continuous environments (VLN-CE [58]) using simulators like Habitat [69]. This introduces greater complexity, as agents must perform mid-level actions such as moving forward or rotating, rather than teleporting between nodes. To bridge the gap between discrete and continuous navigation, some approaches [44, 49, 51, 94] use simulator pre-trained waypoint models [42, 43] that predict candidate positions around the agent and have shown significant performance gains. However, they often struggle to generalize due to their reliance on simulator-specific data. Additionally, the candidate positions predicted by these models only cover nearby locations and do not account for low-level motion planning or obstacle avoidance. In this paper, we aim to advance VLN toward real-world robotics applications, particularly for challenging-legged robots. NaVILA handles both high-level decision-making and generates low-level actions to control the robot's full motion. Additionally, we introduce a new VLN benchmark built on Isaac Sim, offering a more realistic simulation environment, which we believe will benefit future work in VLN.

Robot Foundation Models. Robot foundation models aim to provide a unified framework that processes inputs from various modalities (e.g., vision and language) and directly outputs actions to enable robots to perform complex tasks. Existing works [7, 8, 95] trained on large-scale robotic datasets to get general robot policies, but mainly focusing on manipulation tasks. Doshi et al. [96] and Yang et al. [97] proposed end-to-end visual-language cross-embodiment models for different robotic tasks. Recently, several foundational navigation models have been proposed [98–100]. However, they mainly focus on goal navigation with either short language descriptions or a target image as input. As for legged robots, Ding et al. [101] proposed a unified model to leverage vision and language inputs and generate executable low-level actions. Another line of work [102, 103] focuses on training specialized policies as a skill bank to handle specific actions, with either a VLM or LLM serving as the controller to decide which skill to execute. Similarly, these methods cannot perform instruction-following tasks, as they struggle to comprehend complex instructions that are essential for general navigation. To address this, we propose a VLA model specifically designed for general vision language navigation tasks.

Legged Robot Locomotion Learning. Legged robot locomotion learning focuses on enabling robots to traverse various terrains. Previous works [104, 105] rely solely on robot's proprioceptive information struggle in scenarios like obstacle avoidance. Other end-to-end vision-based approaches [106–109] are vulnerable to extreme environmental conditions, such as intense sunlight, due to the limitations of sensors. Lee et al. [38] incorporate LiDAR sensors in addition to depth cameras to improve terrain sensing, but rely on time-inefficient two-stage training. Additionally, during training, Miki et al. [39] query predefined terrain heights to construct a height map and then rely on an external tool[110] for height map generation during deployment, resulting in discrepancies between training and deployment. To overcome these limitations, we propose a single-stage RL framework that integrates LiDAR sensing inputs during training, allowing the robot to directly learn from interacting with the environments for better efficiency and robustness in complex scenarios.

表 VI：在四足机器人（Unitree Go2）和人形机器人（Booster T1）上于不同环境（工作区、家庭和户外）进行的真实世界实验。简单和复杂分别指简单和复杂的指令遵循任务。注意，† 表示未使用人类导览视频训练的模型。

	工作区				Home				户外			
	简单		复杂		简单		复杂		简单		复杂	
	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑	NE↓	SR↑
宇树 Go2												
GPT-4o [28]	2.01	0.67	2.38	0.33	1.49	0.53	3.00	0.00	-	0.67	-	0.50
NaVILA †	2.00	0.60	1.81	0.73	2.17	0.47	2.32	0.40	-	0.00	-	0.00
NaVILA	**1.29**								-	**1.00**	-	**0.83**
Booster T1												
GPT-4o [28]	1.53	0.67	2.78	0.13	-	-	-	-	-	0.44	-	-
NaVILA †	2.36	0.40	2.16	0.62	-	-	-	-	-	0.22	-	-
NaVILA	**1.18**				-	-	-	-	-	**0.89**	-	-

结果表明，借助人类视频，模型能够更好地泛化到户外场景，并在所有环境中取得更高的成功率。为展示我们两级方法的灵活性，我们还在 Booster Dynamics T1 人形机器人上进行了评估，使用相同的 VLA 模型且无需任何重新训练。尽管存在摄像头高度和摄像头视角变化等差异，NaVILA 仍始终优于基线，凸显了我们模型强大的泛化能力。我们的定性结果见图1和图7。在图7中，我们展示了与语音识别的集成，从而可通过我们的框架实现语音控制导航。这些结果凸显了 NaVILA 在弥合视觉-语言理解与真实世界导航任务之间差距方面的有效性。

IV. 相关工作

视觉导航。视觉导航长期以来一直是机器人学中的研究课题 [71–74]。经典方法依赖于预计算的 [75] 或由深度传感器 [76]或单目相机在定位机器人（SLAM）时构建的几何地图 [77, 78]。最近，使用模仿学习 [79, 80] 和强化学习 [81, 82] 的基于学习的方法已展现出强大性能，并扩展应用到视觉语言导航。

视觉-语言导航。视觉-语言导航（VLN）是具身人工智能中的一个基本挑战，其中智能体使用视觉线索和自然语言指令在复杂环境中导航。该领域随着时间的推移已显著发展。早期研究 [1, 30, 83] 专注于在模拟环境（如 MP3D [84],）中的离散导航，其中智能体在导航图 [31, 85–91] 上的预定义节点之间传送。随着基础模型的进步，许多VLN系统通过利用大规模预训练模型 [25, 92] 和预训练技术 [24,52, 93]，而大幅改进，在此设置中接近人类水平的表现。然而，这种设置强调高层决策，而忽略了底层运动控制的挑战。最近，研究 [12, 54–57]已转向连续环境（VLN-CE [58]），使用像Habitat [69]这样的模拟器。这引入了更大的复杂性，因为智能体必须执行中级动作，例如向前移动或旋转，而不是在节点之间传送。为了弥合离散导航和连续导航之间的差距，一些方法 [44, 49, 51, 94] 使用模拟器预训练的航点模型 [42, 43]，这些模型预测智能体周围的候选位置，并已显示出显著的性能提升。然而，由于它们依赖模拟器特定的数据，往往难以泛化。此外，这些模型预测的候选位置仅覆盖附近位置，并未考虑底层运动规划或避障。在本文中，我们旨在将VLN推向现实世界的机器人应用，特别是对于具有挑战性的腿式机器人。

NaVILA 处理高层决策并生成底层动作以控制机器人的完整运动。此外，我们引入了一个基于 Isaac Sim 构建的新 VLN 基准，提供更真实的模拟环境，我们相信这将有益于 VLN 的未来工作。**机器人基础模型**。 机器人基础模型旨在提供一个统一框架，处理来自各种模态（例如视觉和语言）的输入，并直接输出动作，使机器人能够执行复杂任务。现有工作 [7, 8, 95] 在大规模机器人数据集上训练以获得通用机器人策略，但主要关注操作任务。Doshi 等人 [96] 和 Yang 等人 [97] 提出了针对不同机器人任务的端到端视觉-语言跨具身模型。最近，已经提出了几个基础导航模型 [98–100]。然而，它们主要关注目标导航，输入要么是简短的语言描述，要么是目标图像。对于腿式机器人，Ding 等人 [101]提出了一个统一模型，利用视觉和语言输入并生成可执行的底层动作。另一条工作线 [102, 103] 专注于训练专用策略作为技能库来处理特定动作，由 VLM 或 LLM 充当控制器来决定执行哪个技能。类似地，这些方法无法执行指令跟随任务，因为它们难以理解对通用导航至关重要的复杂指令。为了解决这个问题，我们提出了一个专门为通用视觉语言导航任务设计的 VLA 模型。

腿足式机器人运动学习。 腿足式机器人运动学习旨在使机器人能够穿越各种地形。先前的工作 [104, 105] 仅依赖机器人本体感知信息，在避障等场景中表现不佳。其他基于视觉的端到端方法 [106–109] 由于传感器限制，容易受到极端环境条件（如强烈阳光）的影响。Lee 等人 [38] 在深度相机之外引入 LiDAR 传感器以改善地形感知，但依赖耗时低效的两阶段训练。此外，在训练期间，Miki 等人 [39] 查询预定义地形高度以构建高度图，然后在部署期间依赖外部工具 [110] 生成高度图，导致训练与部署之间存在差异。为克服这些限制，我们提出一种单阶段 RL 框架，在训练期间集成 LiDAR 传感输入，使机器人能够直接从与环境交互中学习，从而在复杂场景中获得更好的效率和鲁棒性。

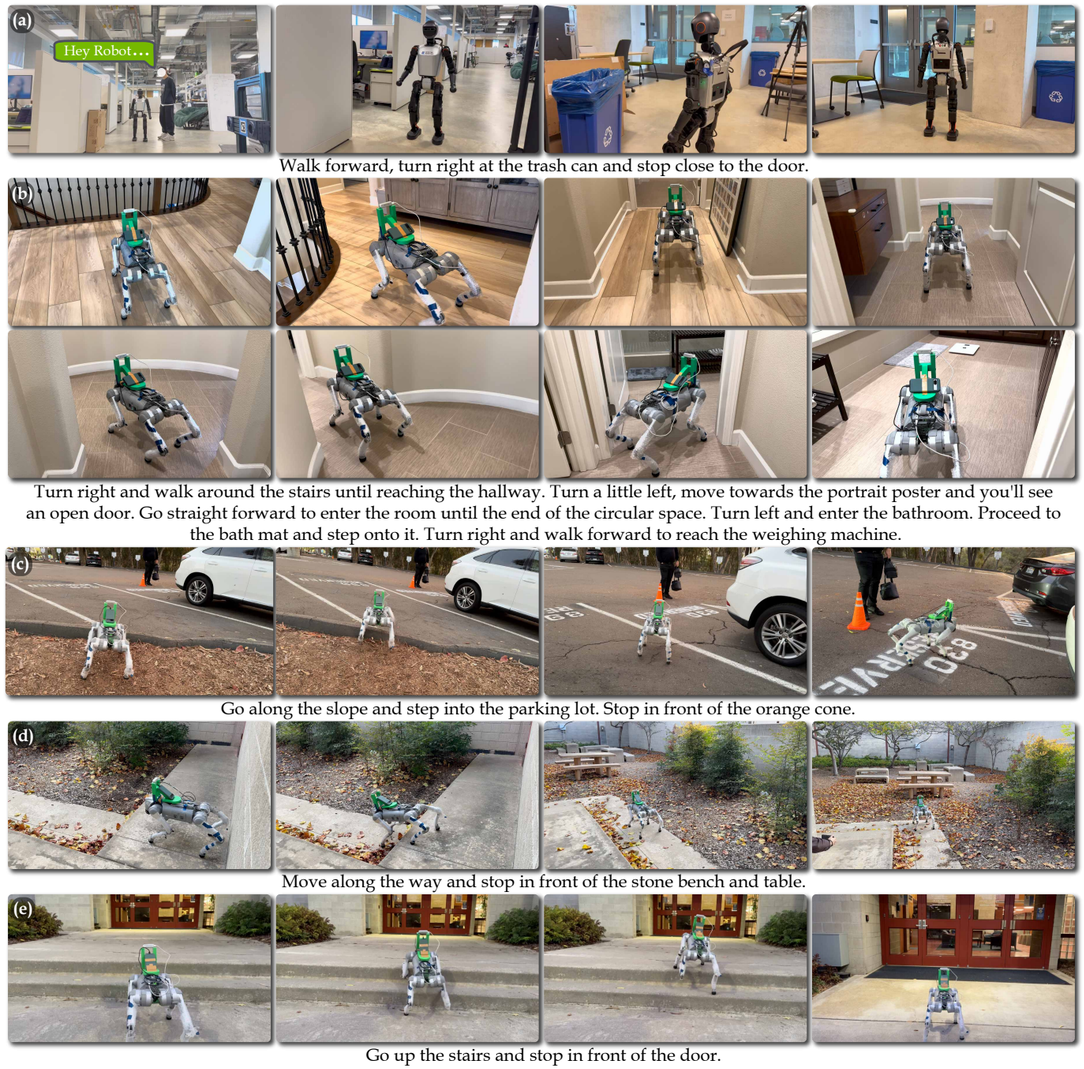

Fig. 7: Qualitative results from the real-world deployment of NaVILA. (a) We integrate speech recognition [70] into NaVILA, allowing a human to control the robot using voice commands that begin with "`Hey Robot!`". (b) The robot successfully handles long-horizon navigation tasks. Given a lengthy instruction, it moves through different areas of the house and stops at the specified goal. (c), (d), and (e) The robot demonstrates its ability to navigate through obstacles, traverse challenging terrains, and climb up and down stairs.

V. Conclusion and Limitations

We introduce NaVILA, a two-level framework that unifies VLAs with locomotion skills for generic navigation. NaVILA generates high-level language commands while a real-time locomotion policy handles obstacle avoidance, enhancing robustness across robots. This design preserves reasoning, prevents overfitting, and enables direct learning from human videos for better generalization. NaVILA achieves a 17% gain on classic VLN benchmarks, outperforms distillation-based low-level policies, surpasses vision-blind policies on VLN-CE-Isaac1K, and demonstrates strong real-world performance across diverse environments and legged robots.

Limitations. While NaVILA demonstrates strong performance, it fails in some real-world cases (see Appx. Sec. E). Enhancing generalizability and spatial understanding through larger-scale training in realistic simulations could help. Additionally, image-based vision-language models are computationally intensive. Advances in long-context LLMs may alleviate this by enabling more efficient sequence processing.

图 7：NaVILA 在真实世界部署中的定性结果。(a) 我们将语音识别 [70] 集成到 NaVILA 中，使人类能够使用以“嘿，机器人！”开头的语音命令控制机器人。(b) 机器人成功处理长时程导航任务。给定一段冗长的指令，它会在房屋的不同区域中移动，并停在指定目标处。(c)、(d) 和 (e) 机器人展示了其穿越障碍物、通过具有挑战性的地形以及上下楼梯的能力。

V. 结论与局限性

我们提出 NaVILA，一个两级框架，将 VLA 与运动技能统一用于通用导航。NaVILA 生成高级语言命令，而实时运动策略处理避障，从而增强跨机器人的鲁棒性。这种设计保留了推理能力、防止过拟合，并能够直接从人类视频中学习，以获得更好的泛化能力。NaVILA 在经典 VLN 基准上取得 17% 的提升，优于基于蒸馏的低层策略，在 VLN- 上超越视觉盲策略 CE-Isaac1K，并在多样化环境和腿式机器人上展现出强大的真实世界性能。

局限性。尽管 NaVILA 表现出强大的性能，但在一些真实世界案例中仍会失败（见附录第 E 节）。通过在逼真仿真中进行更大规模训练来增强泛化能力和空间理解，可能会有所帮助。此外，基于图像的视觉-语言模型计算开销很大。长上下文 LLM 的进步或许能通过实现更高效的序列处理来缓解这一问题。

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A. More Ablation Studies

1) Different Simulation Data Blends: We perform an ablation study to assess the impact of different simulation data blends on training VLA. As shown in Table VII, training navigation data without label rebalancing leads to a significant drop in performance. Additionally, training VLA exclusively on RxR data demonstrates reasonable cross-dataset performance on R2R-CE, supporting our observations in Table II. Lastly, we investigate whether excluding RxR dataset degrades R2R-CE performance. The results suggest that the RxR dataset does not significantly contribute to the R2R-CE performance.

TABLE VII: Results on R2R-CE using different data blends.

	R2R-CE Val Unseen			
	NE ↓	OSR ↑	SR ↑	SPL ↑
NaVILA † (w/o Label Balancing)	7.82	47.5	30.0	25.1
NaVILA † (w/ RxR only)	7.57	40.8	31.5	27.8
NaVILA † (w/o RxR)	6.11	57.0	47.7	42.4
NaVILA †	**5.37**	**57.6**	**49.7**	**45.5**

2) Human Touring Video Data: We perform ablation studies in simulations to assess the effect of using real-world data from YouTube human touring videos. As shown in Table VIII, incorporating this data leads to significant performance gains, with approximately 5% improvements in OS, SR, and SPL metrics. Similarly, results from real-world experiments (Table VI) demonstrate higher success rates and fewer navigation errors. These results validate the effectiveness of our data pipeline and highlight the scalability of NaVILA's framework, which supports easy integration of data from diverse sources.

TABLE VIII: Results on R2R-CE using additional real data from human touring videos.

	R2R-CE Val Unseen			
	NE ↓	OSR ↑	SR ↑	SPL ↑
NaVILA †	5.37	57.6	49.7	45.5
NaVILA	**5.22**	**62.5**	**54.0**	**49.0**

† indicates models trained without human touring videos.

3) Different Memory Sizes: We conduct an ablation study to evaluate the impact of memory size (number of history frames) on the navigation task using R2R-CE benchmark. The results in Table IX show that for R2R-CE, 8 frames are sufficient to cover most instruction horizons, with limited performance gains from increasing the memory size. For real-world experiments, we use an 8-frame memory size due to latency constraints.

TABLE IX: Ablation study on different memory size using R2R-CE [29] Validation Unseen split.

	R2R-CE Val Unseen			
	NE ↓	OSR ↑	SR ↑	SPL ↑
NaVid [12]	5.47	49.0	37.0	35.0
NaVILA † (8 frames)	**5.37**	57.6	49.7	**45.5**
NaVILA † (16 frames)	5.63	55.8	48.6	44.4
NaVILA † (32 frames)	5.74	55.9	49.5	44.1
NaVILA † (64 frames)	5.63	**60.5**	**50.1**	45.4

B. More Qualitative Results

1) VLN-CE-Isaac: Here we show a visualization example highlighting why the Go2 vision policy significantly outperforms the blind policy. As demonstrated in Figure 8, when encountering an obstacle, the VLA, which is not specifically trained for obstacle avoidance, failed to navigate around it effectively. The blind policy, following the VLA's commands without additional sensory input, became stuck at the obstacle. In contrast, the vision-based policy, trained to handle obstacles using LiDAR input, can autonomously avoid dangers even when the high-level VLA model does not detect them.

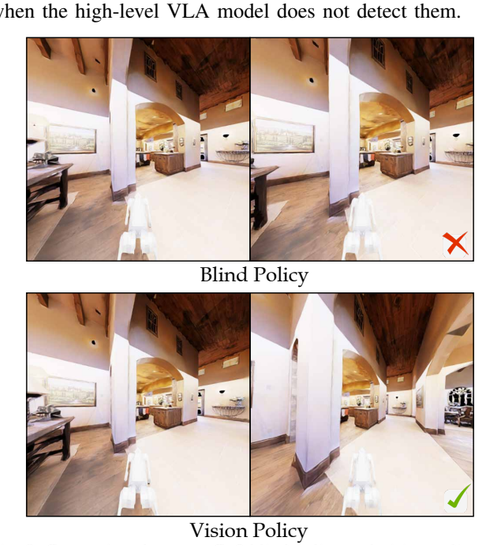

Fig. 8: Comparison between Go2 blind policy and vision policy. The blind policy failed to avoid the obstacles and got stuck. The vision policy detected the obstacle and got around to avoiding it.

2) Real-World: We show more real-world results in Figure 9. NaVILA demonstrates robust performance and exceptional generalization across diverse settings.

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A. 更多消融实验

1) 不同模拟数据混合：我们进行了一项消融研究，以评估不同模拟数据混合对训练 VLA 的影响。如表 VII 所示，在不进行标签重平衡的情况下训练导航数据会导致性能显著下降。此外，仅在 RxR 数据上训练 VLA 在 R2R-CE 上展现出合理的跨数据集性能，这支持了我们在表 II 中的观察。最后，我们研究了排除 RxR 数据集是否会降低 R2R-CE 性能。结果表明，RxR 数据集对 R2R-CE 性能并没有显著贡献。

表 VII：在 R2R-CE 上使用不同数据混合的结果。

	R2R-CE 验证集（未见）			
	NE ↓	OSR ↑	SR ↑	SPL ↑
NaVILA †（无标签平衡）	7.82	47.5	30.0	25.1
NaVILA †（仅使用 RxR）	7.57	40.8	31.5	27.8
NaVILA †（无 RxR）	6.11	57.0	47.7	42.4
NaVILA †	**5.37**	**57.6**	**49.7**	**45.5**

2) 人类游览视频数据：我们在仿真中进行消融研究，以评估使用来自 YouTube 人类游览视频的真实世界数据的效果。如表 VIII，加入这些数据可带来显著的性能提升，在 OS、SR 和 SPL 指标上约有 5% 的提升。同样，真实世界实验（表 VI）的结果表明成功率更高，导航错误更少。这些结果验证了我们数据管道的有效性，并凸显了 NaVILA 框架的可扩展性，该框架支持轻松集成来自多样化来源的数据。

表 VIII：在 R2R-CE 上使用来自人类游览视频的额外真实数据的结果。

	R2R-CE 验证集（未见环境）			
	NE ↓	OSR ↑	SR ↑	SPL ↑
NaVILA †	5.37	57.6	49.7	45.5
NaVILA	**5.22**	**62.5**	**54.0**	**49.0**

†表示未使用人类游览视频训练的模型。

3) 不同记忆大小：我们进行消融研究，以评估记忆大小（历史帧数量）对使用 R2R-CE 基准的导航任务的影响。表 IX 显示，对于 R2R-CE，8 帧足以覆盖大多数指令范围，增加记忆大小带来的性能提升有限。在真实世界实验中，由于延迟限制，我们使用 8 帧的记忆大小。

表 IX：使用 R2R-CE 在不同记忆大小上的消融研究 [29]验证集未见过划分。

	R2R-CE 验证集未见过			
	NE ↓	OSR ↑	SR ↑	SPL ↑
NaVid [12]	5.47	49.0	37.0	35.0
NaVILA †（8 帧）	**5.37**	57.6	49.7	**45.5**
NaVILA †（16 帧）	5.63	55.8	48.6	44.4
NaVILA †（32 帧）	5.74	55.9	49.5	44.1
NaVILA †（64 帧）	5.63	**60.5**	**50.1**	45.4

B. 更多定性结果

1) VLN-CE-Isaac：这里我们展示一个可视化示例，突出说明为什么 Go2 视觉策略显著优于盲策略。如图8所示，当遇到障碍物时，并未专门针对避障训练的 VLA 未能有效绕过障碍物。盲策略在没有额外感官输入的情况下遵循 VLA 的指令，结果卡在障碍物处。相比之下，基于视觉的策略经过训练，可利用 LiDAR 输入处理障碍物，即使高层 VLA 模型未检测到危险，也能自主避开危险。

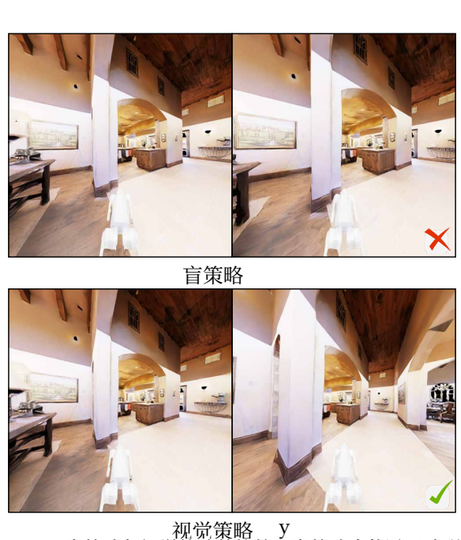


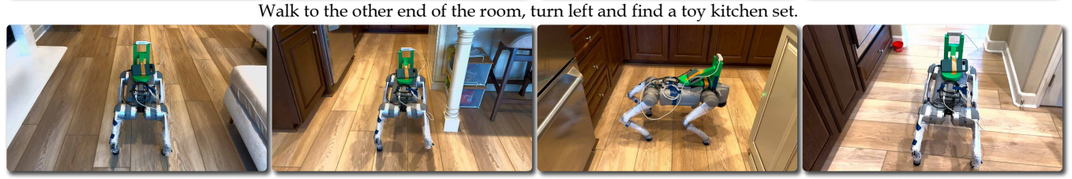
图 8：Go2 盲策略与视觉策略的比较。盲策略未能避开障碍物并卡住了。视觉策略检测到障碍物，并绕行避开它。

2) 真实世界：我们在图9中展示了更多真实世界结果。NaVILA 在多样化设置中展现出稳健的性能和卓越的泛化能力。

Turn left a little. Go straight forward. Stop when you step on the grass.

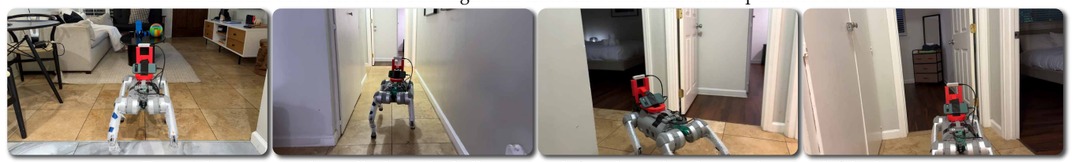
Walk forward. Step on the grass and continue going forward.
Stop when you are close to the big bear statue.

Walk to the other end of the room, turn left and find a toy kitchen set.



Move forward into the kitchen. Turn right at the corner, and then walk straight forward. Stop in front of the red bowl.

Walk forward out of the room. Turn right and enter the other room and stop in front of the table.



Turn right immediately, walk along the hallway, turn left at the end and enter the most left bedroom.

Walk all the way down, turn left at the intersection and find a bookshelf.

Walk all the way down, turn left at the intersection and find a ball.

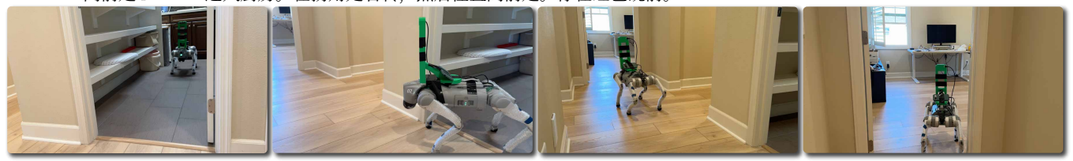
Fig. 9: We show more results in diverse environments, such as urban streets, campus sidewalks, courtyards, and different houses. These settings add significant variety and challenges, including challenging terrains, dynamic objects, and different lighting conditions. The results NaVILA achieved represent a significant milestone, showcasing capabilities that have never been demonstrated before.

稍微向左转。径直向前走。踩到草地时停下。

向前走。踩上草地后继续向前走。靠近大熊雕像时停下。

走到房间的另一端，左转，找到一个玩具厨房套装。

向前走 i　　进入厨房。在拐角处右转，然后径直向前走。停在红色碗前。



向前走出房间。右转进入另一个房间，停在桌子前面。

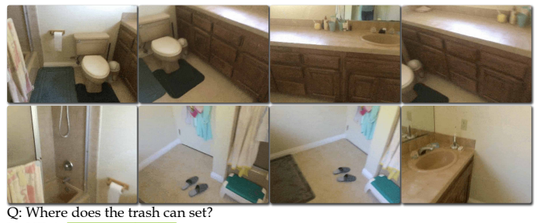
立即右转，沿着走廊前行，在尽头左转，然后进入最左边的卧室。

一直往前走　　到尽头，在十字路口左转，找到一个书　　架。

一直走到尽头，在十字路口左转，找到一个球。

图9：我们展示了在多样环境中的更多结果，例如城市街道、校园人行道、庭院以及不同的房屋。这些场景带来了显著的多样性和挑战，包括具有挑战性的地形、动态物体以及不同的光照条件。NaVILA 取得的成果代表了一个重要的里程碑，展示了此前从未展现过的能力。

3) Spatial Scene Understanding: In Figure 10, we show NaVILA's qualitative results on spatial scene understanding using the ScanQA benchmark. Given a sequence of images sampled from a video, NaVILA can ground and locate objects correctly.



Q: Where does the trash can set?
NaVILA: To right of toilet.
Q: What is hanging on the wall between the toilet and the bath tub?
NaVILA: Toilet paper.

Q: What can be seen on the wall on the right of the chair?
NaVILA: Whiteboard.
Q: What color is the mini fridge in the corner of the room?
NaVILA: Black.

Fig. 10: Spatial scene understanding results on ScanQA benchmark.

C. More Implementation Details

1) Video Navigation Trajectory Summarization: We provide the data prompts for our auxiliary task of video navigation trajectory summarization. Following the approach in [12], we construct prompt templates that characterize the LLM as a robot designed for navigation. We process the trajectory videos into history frames, insert the frame tokens into the prompt, and ask the LLM to infer the navigation instructions from the video. This task is designed to enhance the robot's scene understanding and its familiarity with the instruction format.

```
Assume you are a robot designed for navigation. You
are provided with captured image sequences:

<frame3><frame6><frame9>...

Based on this image sequence, please describe the
navigation trajectory of the robot.
```

2) VLA Hyperparameters: Please refer to VILA's paper for details on the hyperparameters used in the first two stages. In the instruction fine-tuning stage, we use a learning rate of $1e^{-4}$ with cosine decay and a warm-up ratio of 0.03. We will release both our training code and data upon paper publication.

3) Locomotion Motion Policy: The reward functions and domain randomization used during Go2 locomotion policy training are listed in Table X and Table XI. The robust policy is trained on flat, rough, slope and obstacle terrains shown in Figure 11. LiDAR and height map settings are detailed in Table XII.

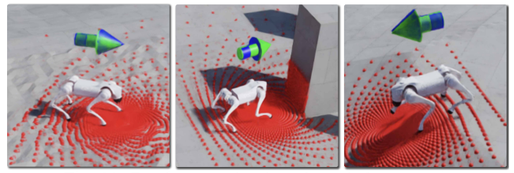


Fig. 11: Random rough, obstacle and slope terrain.

TABLE X: Reward function parameters for training RL policy.

Reward	Expression	Weight
Linear velocity tracking	$\exp(-\lVert v_{xy}^{\text{cmd}} - v_{xy} \rVert_2^2)$	1.5
Angular velocity tracking	$\exp(-\left(\omega_{\text{yaw}}^{\text{cmd}} - \omega_{\text{yaw}}\right)^2)$	1.5
Linear velocity penalty (z)	v_z^2	-2.0
Angular velocity penalty (xy)	$\lVert \boldsymbol{\omega}_{xy} \rVert_2^2$	-0.05
Flat orientation	$\lVert \mathbf{g} \rVert_2^2$	-2.0
Joint accelerations	$\lVert \ddot{\boldsymbol{\theta}} \rVert^2$	-2.5×10^{-7}
Energy	$-\lVert \tau \dot{q} \rVert_2^2$	-2×10^{-5}
Body height	$(h^{\text{target}} - h)^2$	-5.0
Feet slipping	$-\lVert v_{\text{feet}} \cdot \mathbf{1}[F_{\text{feet}} > 1] \rVert_2$	0.05

TABLE XI: Domain randomization parameters for training RL policy.

Parameter	Value
Body Mass	[-3.0, 3.0]
Static Ground Friction	[0.4, 4.0]
Dynamic Ground Friction	[0.4, 4.0]
Motor Strength	[0.9,1.1]
System Delay	$[\Delta_t, \Delta_t]$

TABLE XII: LiDAR and Height Map parameters in simulation.

Parameter	Value
Channels	32
Vertical Range (degrees)	(0, 90)
Horizontal Range (degrees)	(-180, 180)
Horizontal Resolution (degrees)	4
Voxel Size (m)	0.06
X Range (m)	[-0.8, 0.2]
Y Range (m)	[-0.8, 0.8]
Z Range (m)	[0.05, 0.5]

4) VLA Compute Resources: The first two stages of NaVILA are inherited from VILA [13], which is trained on 16 A100 GPU nodes, with each node having 8 GPUs. The training times for each stage of our 8B model are as follows: connector initialization takes 4 hours, visual language pre-training takes 30 hours. The final visual instruction-tuning stage is experimented on 4 A100 GPU nodes, taking 18 hours. During inference time, the VLA model in NaVILA can be served using a single RTX 4090 GPU with roughly 1 FPS.

D. Parameter-efficient Quantization

We explore optimization techniques to improve NaVILA's inference efficiency. Specifically, we apply AWQ [111], a state-of-the-art quantization method for VLMs, to the FP16 NaVILA-8B model. By converting it to the W4A16 format (low-bit weight-only quantization), we achieved significant improvements: memory requirements dropped by half, and processing speed improved by about 40%. Most importantly, navigation capabilities remained robust. Results are detailed in Table XIII. These optimizations make NaVILA deployable directly on the robot, which will significantly eliminate image transmission time. We leave this as future work.

3）空间场景理解：在图10中，我们展示了NaVILA 在空间场景理解上使用 ScanQA 基准的定性结果。给定从视频中采样得到的一系列图像，NaVILA 能够正确地对物体进行 grounding 和定位。

问：垃圾桶放在哪里？NaVILA：　　问：马桶和浴缸之间的墙上挂着什么？NaVILA：

问：椅子右边的墙上可以看到什么？NaVILA：　　问：房间角落里的迷你冰箱是什么颜色？NaVILA：

图 10：ScanQA 基准上的空间场景理解结果。

C. 更多实现细节

1) 视频导航轨迹摘要：我们提供用于视频导航轨迹摘要这一辅助任务的数据提示。遵循 [12],中的方法，我们构建提示模板，将 LLM 描述成专为导航设计的机器人。我们将轨迹视频处理为历史帧，将帧标记插入提示中，并要求 LLM 从视频中推断导航指令。此任务旨在增强机器人的场景理解能力及其对指令格式的熟悉程度。

假设你是一个专为导航设计的机器人。你将获得捕获的图像序列：

<帧3><帧6><帧9>...

基于该图像序列，请描述机器人的导航轨迹。

*2) VLA*超参数：关于前两个阶段使用的超参数，请参阅 VILA 的论文以了解详情。在指令微调阶段，我们使用的学习率为 $1e^{-4}$，并采用余弦衰减，预热比例为 0.03。我们将在论文发表后发布训练代码和数据。

3) 移动运动策略：Go2 运动策略训练中使用的奖励函数和域随机化列于表 X 和表 XI。鲁棒策略在平坦、崎岖、斜坡和障碍地形上进行训练，如图 11。LiDAR 和高度图设置详见表 XII。

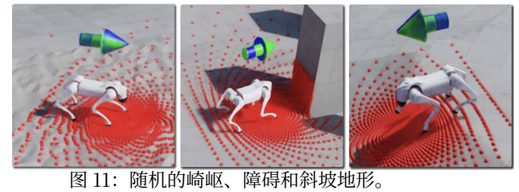


图 11：随机的崎岖、障碍和斜坡地形。

表 X：用于训练 RL 策略的奖励函数参数。

奖励	表达式	权重
线速度跟踪	$\exp(-\lVert v_{xy}^{\text{cmd}} - v_{xy} \rVert_2^2)$	1.5
角速度跟踪	$\exp(-\left(\omega_{\text{yaw}}^{\text{cmd}} - \omega_{\text{yaw}}\right)^2)$	1.5
线速度惩罚 (z)	v_z^2	-2.0
角速度惩罚 (xy)	$\lVert \boldsymbol{\omega}_{xy} \rVert_2^2$	-0.05
水平姿态	$\lVert \mathbf{g} \rVert_2^2$	-2.0
关节加速度	$\lVert \ddot{\boldsymbol{\theta}} \rVert^2$	-2.5×10^{-7}
能量	$-\lVert \tau \dot{q} \rVert_2^2$	-2×10^{-5}
机体高度	$(h^{\text{target}} - h)^2$	-5.0
脚部打滑	$-\lVert v_{\text{feet}} \cdot \mathbf{1}[F_{\text{feet}} > 1] \rVert_2$	0.05

表 XI：用于训练 RL 策略的域随机化参数。

参数	值
机体质量	[-3.0, 3.0]
静态地面摩擦	[0.4, 4.0]
动态地面摩擦	[0.4, 4.0]
电机强度	[0.9,1.1]
系统延迟	$[\Delta_t, \Delta_t]$

表 XII：仿真中的 LiDAR 和高度图参数。

参数	值
通道	32
垂直范围（度）	(0, 90)
水平范围（度）	(-180, 180)
水平分辨率（度）	4
体素大小（米）	0.06
X 范围（米）	[-0.8, 0.2]
Y 范围（米）	[-0.8, 0.8]
Z 范围（米）	[0.05, 0.5]

4) VLA 计算资源：NaVILA 的前两个阶段继承自 VILA [13], ，该系统在 16 个 A100 GPU 节点上训练，每个节点有 8 块 GPU。我们 8B 模型各阶段的训练时间如下：连接器初始化耗时 4 小时，视觉语言预训练耗时 30 小时。最后的视觉指令微调阶段在 4 个 A100 GPU 节点上进行实验，耗时 18 小时。在推理时，NaVILA 中的 VLA 模型可仅使用单块 RTX 4090 GPU 提供服务，速度约为 1 FPS。

D. 参数高效量化

我们探索优化技术，以提升 NaVILA 的推理效率。具体而言，我们将 AWQ [111]，（一种面向 VLM 的最先进量化方法）应用于 FP16 NaVILA-8B 模型。通过将其转换为 W4A16 格式（低比特仅权重量化），我们取得了显著改进：内存需求减半，处理速度提升约 40%。最重要的是，导航能力保持稳健。结果详见表十三。这些优化使 NaVILA 可直接部署在机器人上，这将显著消除图像传输时间。我们将此留作未来工作。

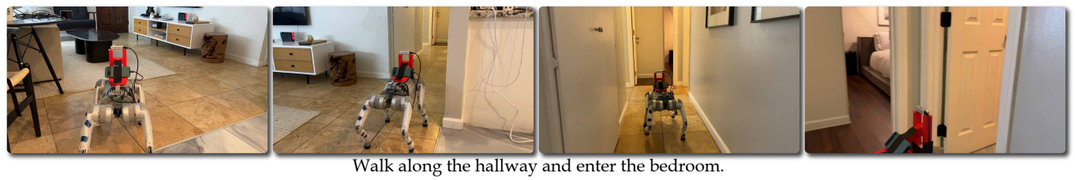
Fig. 12: Obstacle avoidance screenshots. Locomotion policy can ensure collision-free in the face of high grass, certain transparent glass, and large objects under strong sunlight. The policy presents robustness on sand and grass terrains.

TABLE XIII: NaVILA quantization results. The computational cost is tested on RTX 4090 with 1737 context tokens and 10 generated tokens, using a sample from R2R-CE as the test case.

	Computational Cost		R2R Val-Unseen			
	Total Latency (ms) ↓	GPU Memory (GB) ↓	NE ↓	OS ↑	SR ↑	SPL ↑
NaVILA (FP16)	594.58	18.5	5.37	57.6	49.7	45.5
NaVILA (W4A16)	**367.80**	**8.6**	5.66	56.8	48.2	43.6

E. Limitations

In Figure 13, we highlight a failure case in the real world where the robot initially follows the prompt but ultimately fails to reach the bedroom. This failure arises from the robot's inability to perform effective error correction when deviations occur. To further enhance performance, improving generalizability and spatial understanding is key. One potential direction is larger-scale training on more realistic simulations, which could provide more diverse navigation scenarios and error recovery cases. Additionally, incorporating explicit reasoning data during training could help the model better anticipate and correct mistakes.



Walk along the hallway and enter the bedroom.

Fig. 13: Failure case of NaVILA.

图 12：避障截图。运动策略能够确保在面对高草丛、某些透明玻璃以及强阳光下的较大物体时无碰撞。该策略在沙地和草地地形上表现出鲁棒性。

表 XIII：NaVILA 量化结果。计算成本在 RTX 4090 上使用 1737 个上下文 token 和 10 个生成 token 进行测试，并以来自 R2R-CE 的一个样本作为测试用例。

	计算成本		R2R Val-Unseen			
	总延迟 (ms) ↓	GPU 内存 (GB) ↓	NE ↓	OS ↑	SR ↑	SPL ↑
NaVILA (FP16)	594.58	18.5	5.37	57.6	49.7	45.5
NaVILA (W4A16)	**367.80**	**8.6**	5.66	56.8	48.2	43.6

E. 局限性

在图13中，我们重点展示了一个现实世界中的失败案例：机器人最初遵循提示，但最终未能到达卧室。这一失败源于当出现偏差时，机器人无法进行有效的纠错。为了进一步提升性能，提升泛化能力和空间理解至关重要。一个潜在方向是在更逼真的仿真中进行更大规模的训练，这可以提供更多样的导航场景和错误恢复案例。此外，在训练中加入显式推理数据有助于模型更好地预见并纠正错误。

沿走廊行走并进入卧室。

图 13：NaVILA 的失败案例。